AI ETHICS

MAKING TRUST, RESPONSIBILITY AND ALIGNMENT WORK IN PRACTICE

Markus Kneer
Chair of Ethics of AI, Graz University
Research Associate, Digital Society Initiative, University of Zurich

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KEY QUESTION

- Q: Are our strategies for allocating responsibility fit for the world we live in?

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Support us →

Canada

Air Canada ordered to pay customer who was misled by airline's chatbot

Company claimed its chatbot 'was responsible for its own actions' when giving wrong information about bereavement fare



February 2024; https://archive.is/ZAnQf

Waymo

California police stumped after trying to ticket driverless car for illegal U-turn

San Bruno officers pull over Waymo but say a ticket wasn't issued, as 'our citation books don't have a box for "robot"'



October 2025; https://archive.is/9DkVL

KEY QUESTION

- Q: Are our traditional strategies for allocating responsibility fit for the world we live in?
- A: Well, maybe not.

THE PLAN

1. Philosophical reasons for revising responsibility allocation
2. Psychological reasons for revising responsibility allocation
3. New models and what they can do
4. Conclusion (i.e. more questions)

I. PHILOSOPHICAL REASONS

RESPONSIBILITY

RESPONSIBILITY

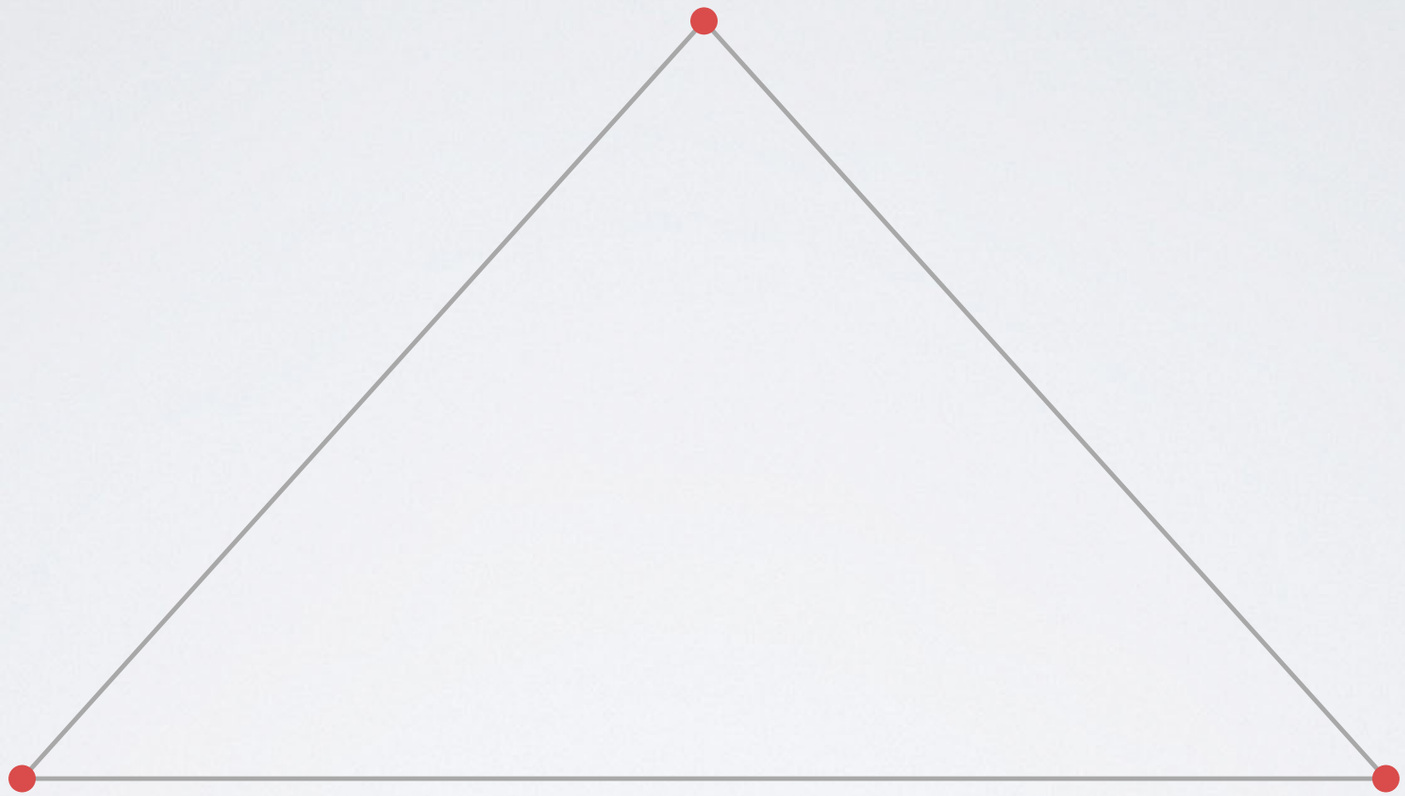
- Roughly: Negative moral responsibiltiy/accountability = blame

- Ethics: Who is morally responsible?
- Law: Who is legally culpable?
- Moral Psychology: Whom do people *consider* responsible?

Shoemaker, D. (2011). Attributability, answerability, and accountability: Toward a wider theory of moral responsibility. *Ethics*

Ethics

Who is morally responsible?

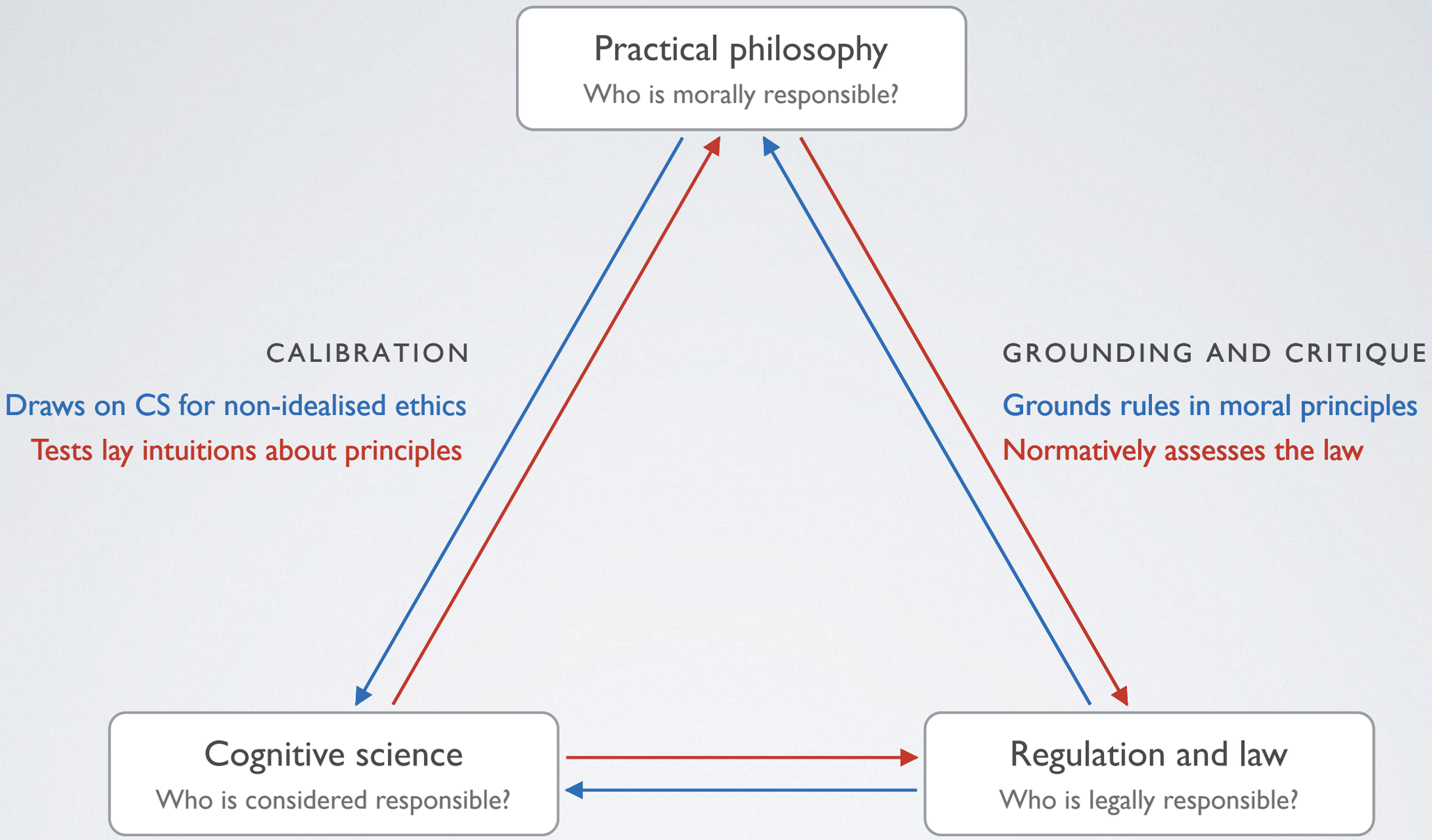


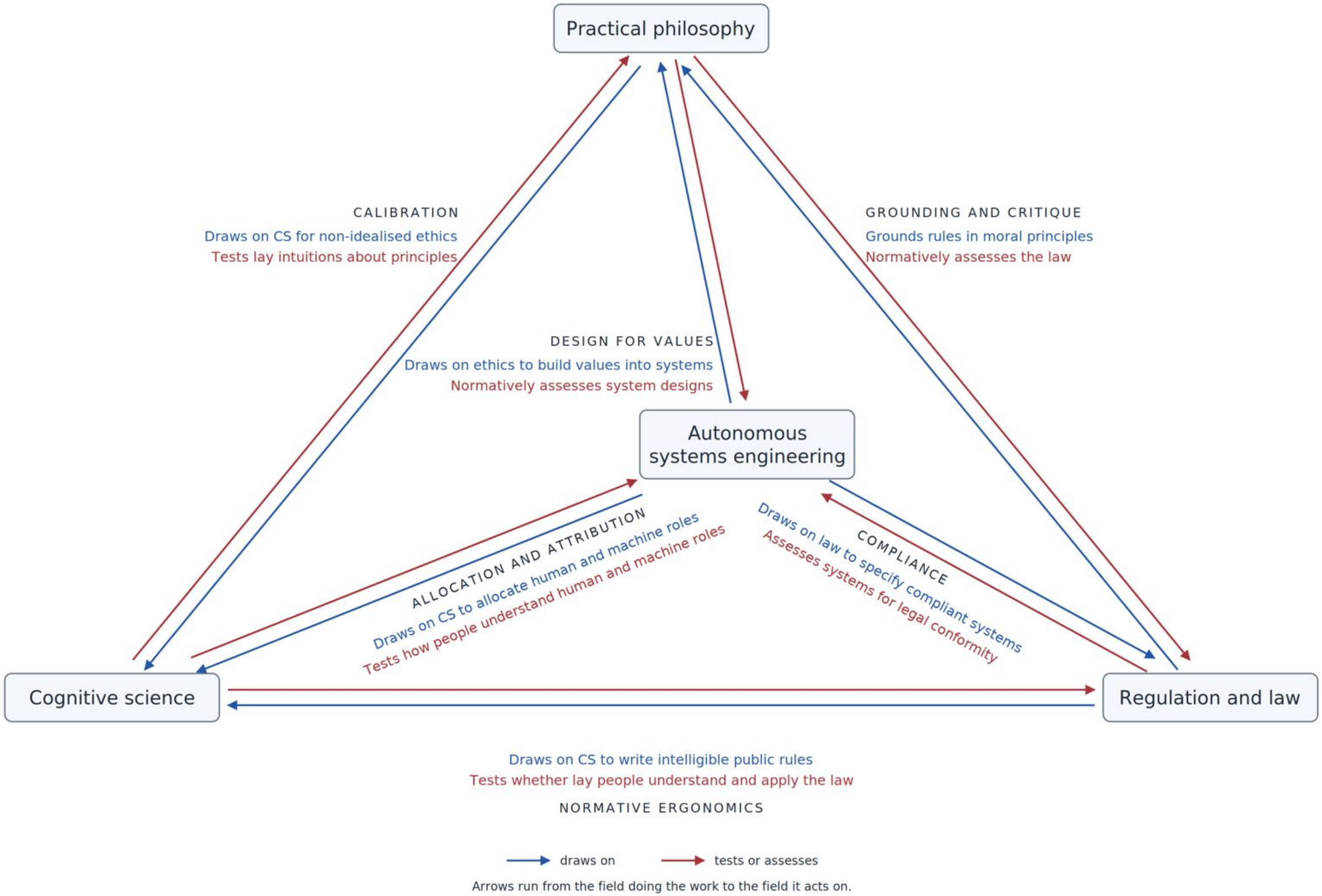
Psychology

Whom do people consider responsible?

Law / Governance

Who is legally responsible?


Practical philosophy
Who is morally responsible?
CALIBRATION
Draws on CS for non-idealised ethics
Tests lay intuitions about principles
GROUNDING AND CRITIQUE
Grounds rules in moral principles
Normatively assesses the law
Cognitive science
Who is considered responsible?
Regulation and law
Who is legally responsible?
Draws on CS to write intelligible public rules
Tests whether lay people understand and apply the law
NORMATIVE ERGONOMICS
draws on
tests or assesses
Arrows run from the field that acts to the field it acts on.


Practical philosophy
CALIBRATION
Draws on CS for non-idealised ethics
Tests lay intuitions about principles
GROUNDING AND CRITIQUE
Grounds rules in moral principles
Normatively assesses the law
DESIGN FOR VALUES
Draws on ethics to build values into systems
Normatively assesses system designs
Autonomous systems engineering
ALLOCATION AND ATTRIBUTION
Draws on CS to allocate human and machine roles
Tests how people understand human and machine roles
COMPLIANCE
Draws on law to specify compliant systems
Assesses systems for legal conformity
Cognitive science
Regulation and law
Draws on CS to write intelligible public rules
Tests whether lay people understand and apply the law
NORMATIVE ERGONOMICS
draws on
tests or assesses
Arrows run from the field doing the work to the field it acts on.

RESPONSIBILITY GAPS

RESPONSIBILITY GAPS

1. Self-learning autonomous systems cannot be responsible.
2. Sometimes, no human agent (programmer, user, owner) can be *justly* held responsible.
3. Hence, "responsibility gaps" can arise: Situations where no-one can be justly held responsible for AI-caused harm.

Matthias (2004). The responsibility gap: Ascribing responsibility for the actions of learning automata. *Ethics and information technology.*

Sparrow, R. (2007). Killer robots. *Journal of applied philosophy.*

RESPONSIBILITY GAPS

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Autonomy of System: Hard to predict & control.

RESPONSIBILITY GAPS

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Autonomy of System

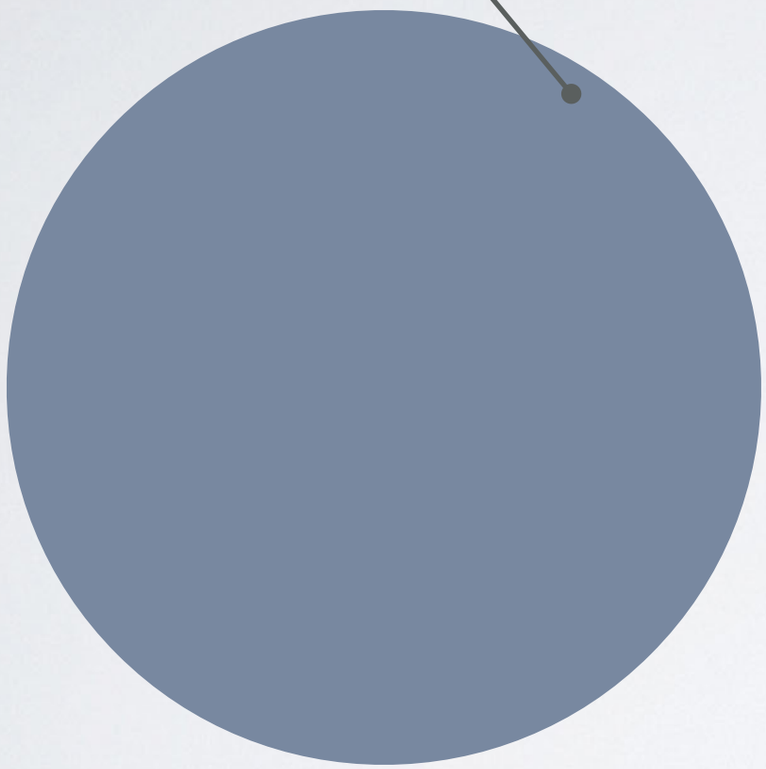
"Many hands problem" - the collective is resp., but no single individual.

RESPONSIBILITY GAPS

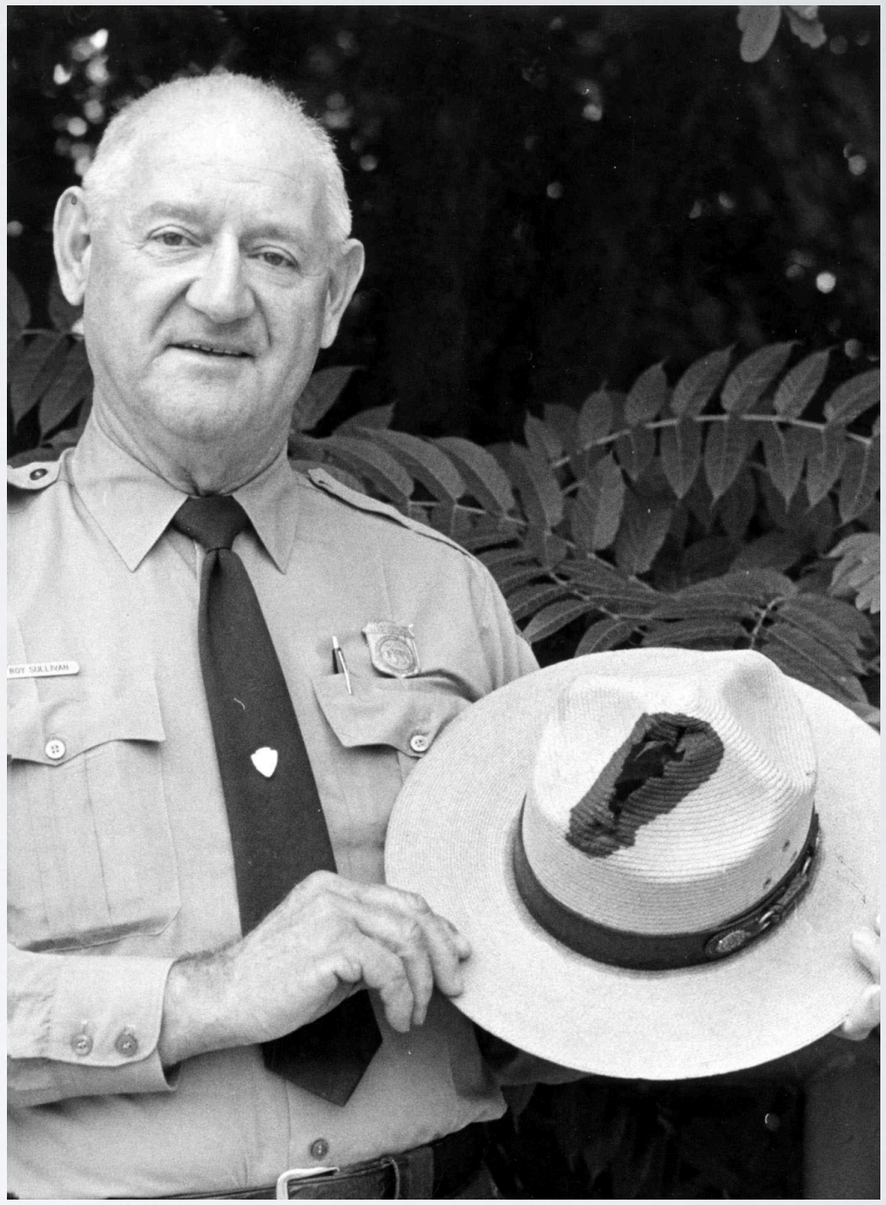
- Doesn't mean that responsibility gaps *always* arise in human-AI interaction - just that they *can* arise.
- Doesn't mean that responsibility gaps *need* to arise, at least not if we change our practices of responsibility attribution.

TYPES OF HARM

Simple harm (tragedy)

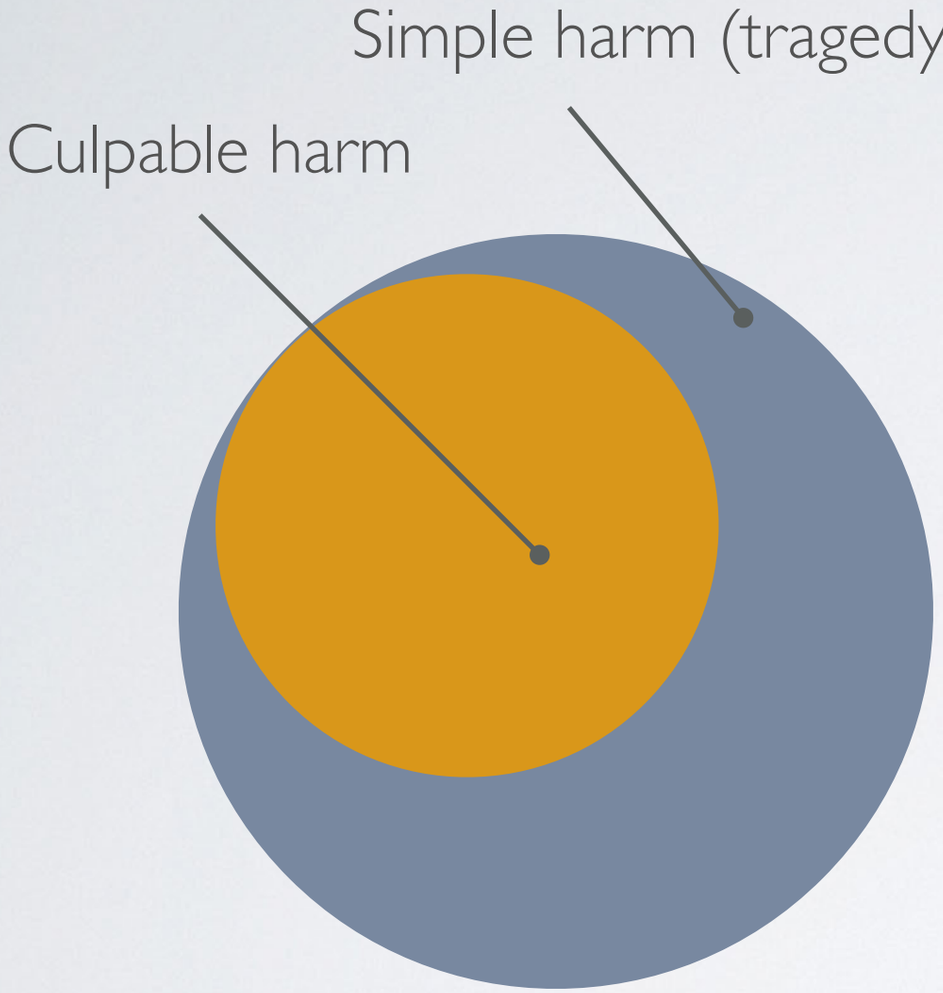


No human-AI interaction

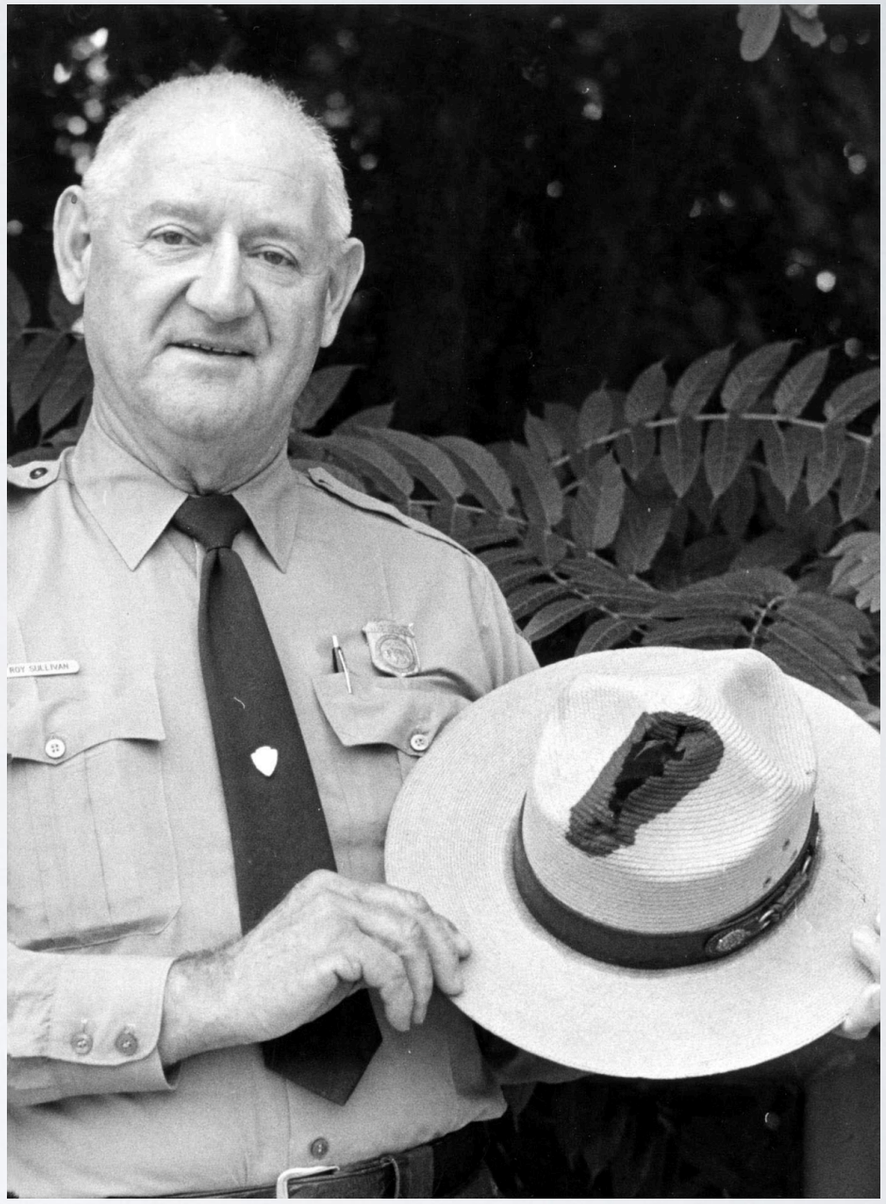


Roy Sullivan (1912-1983), c. mid 1970s,
the "human lightning rod".

TYPES OF HARM

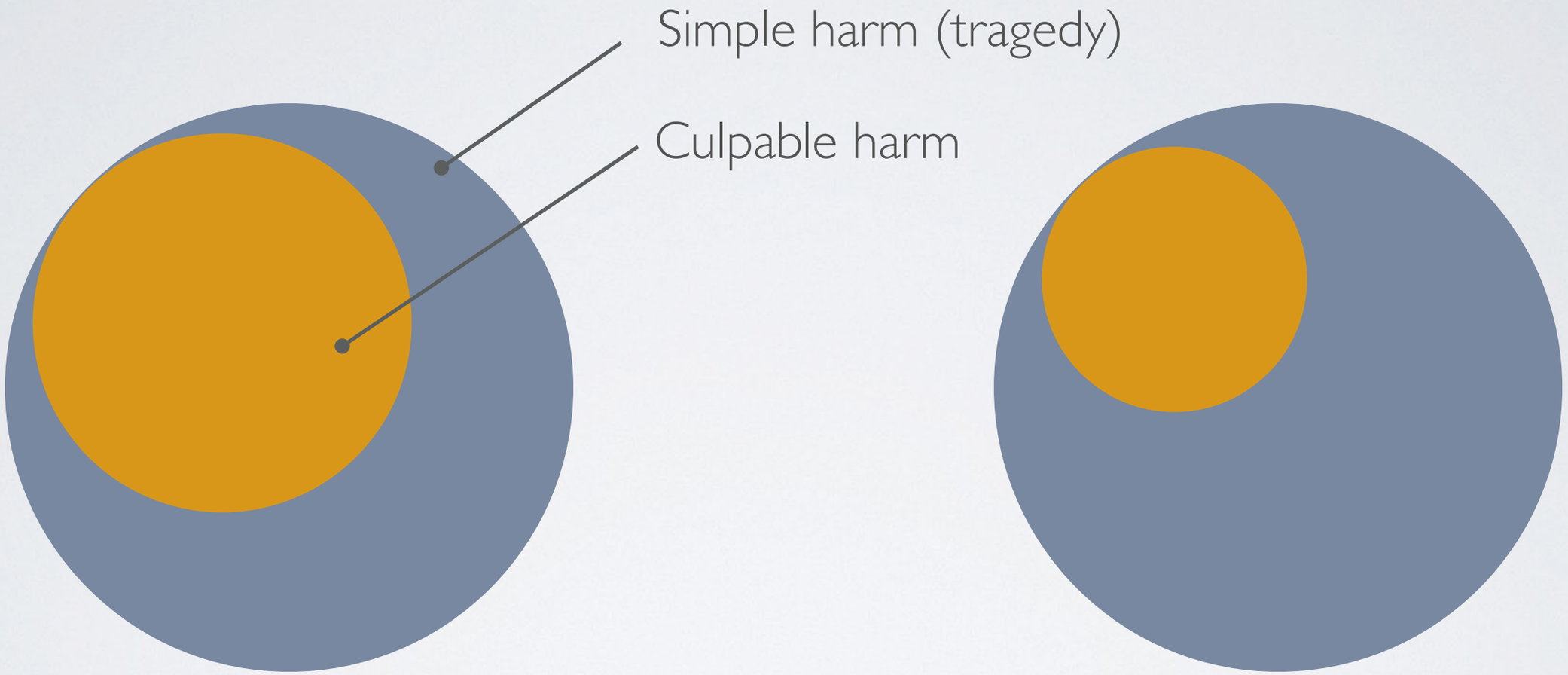


No human-AI interaction



Roy Sullivan (1912-1983), c. mid 1970s, the "human lightning rod".

TYPES OF HARM

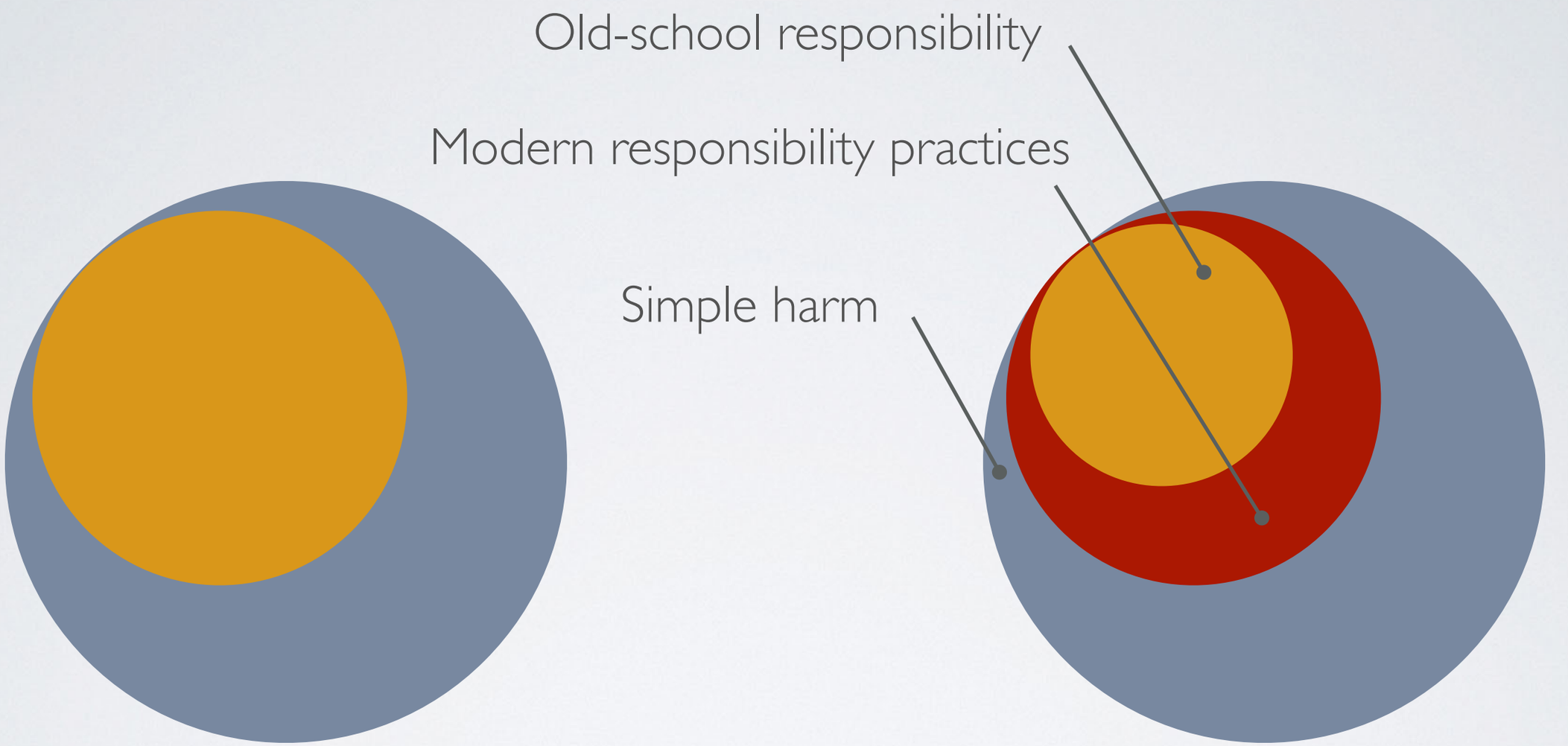


No human-AI interaction

A lot of human AI interaction

Our responsibility frameworks
might be inadequate for the new world.

RESPONSIBILITY GAPS



No human-AI interaction

A lot of human AI interaction

We might need *new* responsibility frameworks.

I. PSYCHOLOGICAL REASONS

KEY QUESTION

- Q: Are our traditional strategies for allocating responsibility fit for the world we live in?
- Philosophy: Well, maybe not - responsibility gaps.
- Psychology: Well, maybe not - people's judgments are all over the place.

THREATS TO JUDGING RESPONSIBILITY

- Blame misattribution
  - Inappropriate inculpation of non-responsible agents (AI, humans).
  - Inappropriate exculpation of responsible humans
- Blame manipulation
  - The deliberate inappropriate inculpation/exculpation, e.g. moral or legal "scapegoating" (e.g. Air Canada)

Danaher (2016). Robots, law and the retribution gap. *Ethics and Information Technology*.

RESPONSIBILITY IN HUMAN-AI INTERACTION

DESIGN

Scenario: A military general deploys a pilot, who bombs a column of surrendering soldiers.



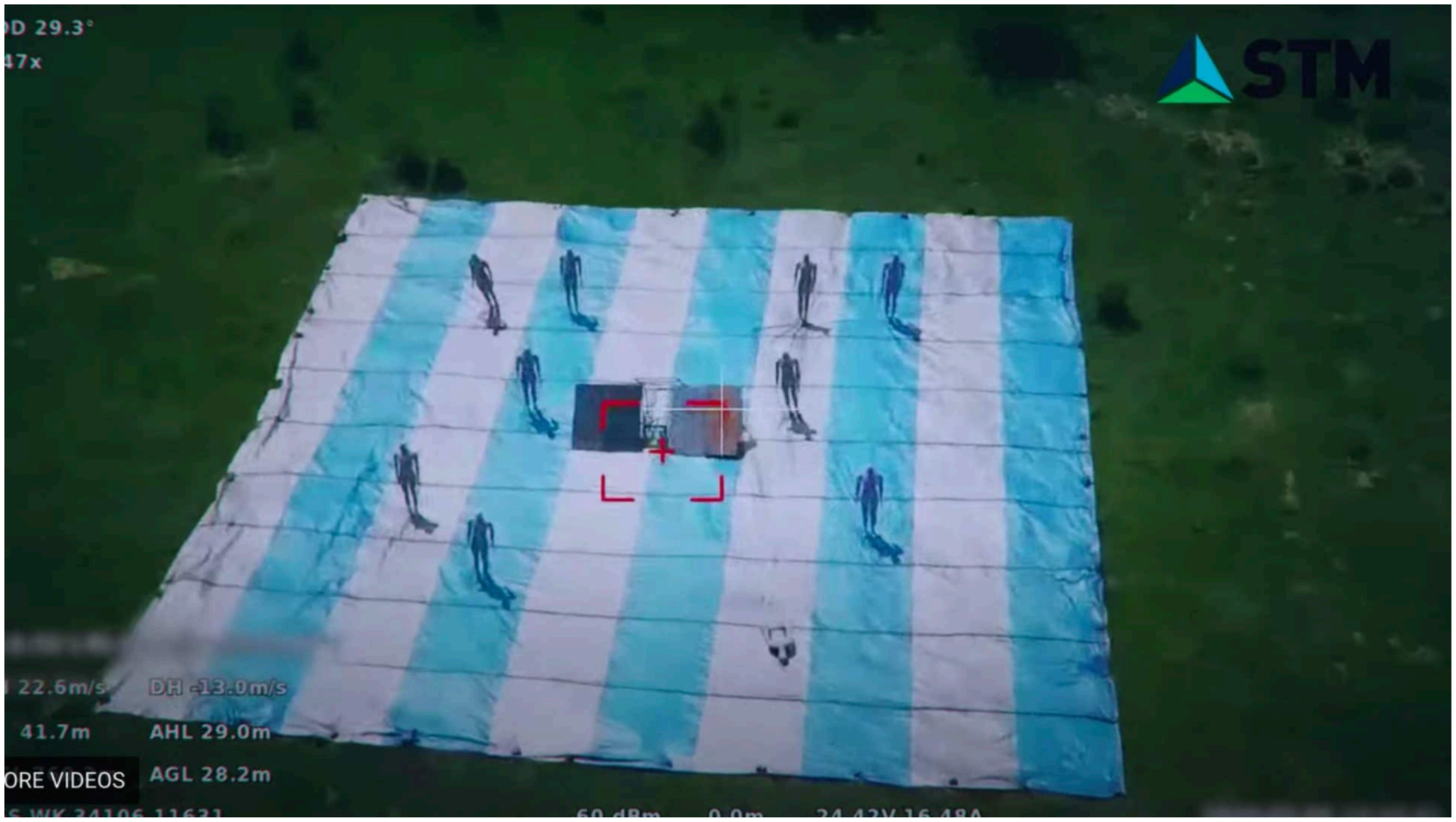
AI-driven autonomous weapon system ("robot")

Human pilot

Sparrow, 2007

Was a flying killer robot used in Libya? Quite possibly

By Zachary Kallenborn | May 20, 2021



A screenshot from a promotional video advertising the Kargu drone. In the video, the weapon dives toward a target before exploding.

https://thebulletin.org/2021/05/was-a-flying-killer-robot-used-in-libya-quite-possibly/

UNITED NATIONS



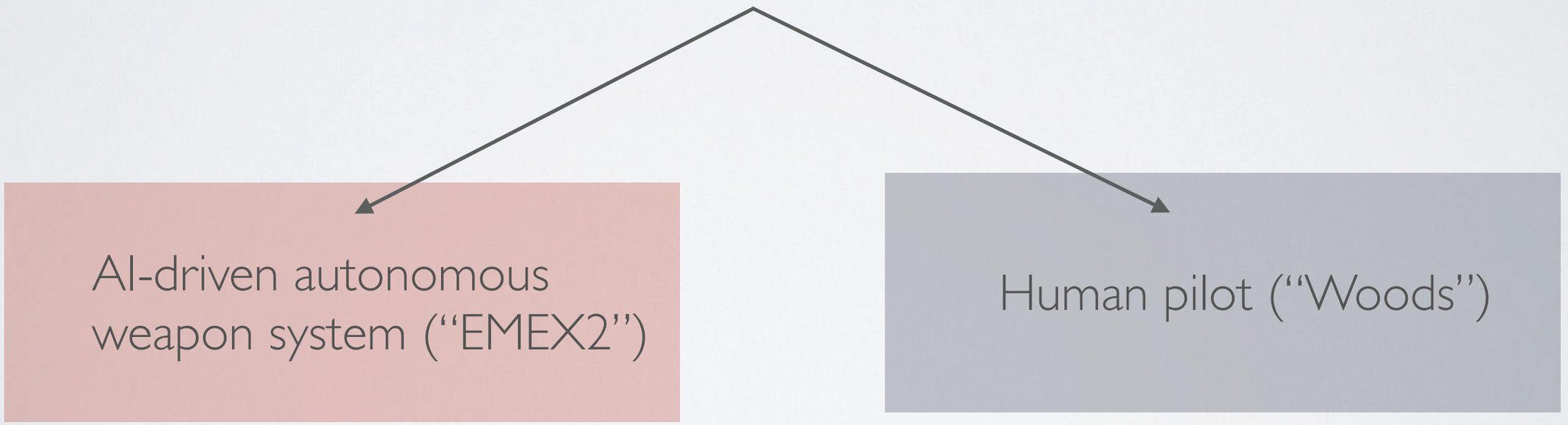
UN Meeting on LAWS, April 2018

CAMPAIGN TO STOP KILLER ROBOTS



DESIGN

Scenario: A military general deploys a pilot, who bombs a column of surrendering soldiers.



Sparrow, 2007

QUESTIONS

Q1: How **morally wrong** do you consider the action of dropping the bomb on the surrendering soldiers?

Q2: To what extent do you consider [Woods/EMEX2] **causally responsible** for the death of the surrendering soldiers?

Q3: To what extent do you consider [Woods/EMEX2] **morally responsible** for the death of the surrendering soldiers?

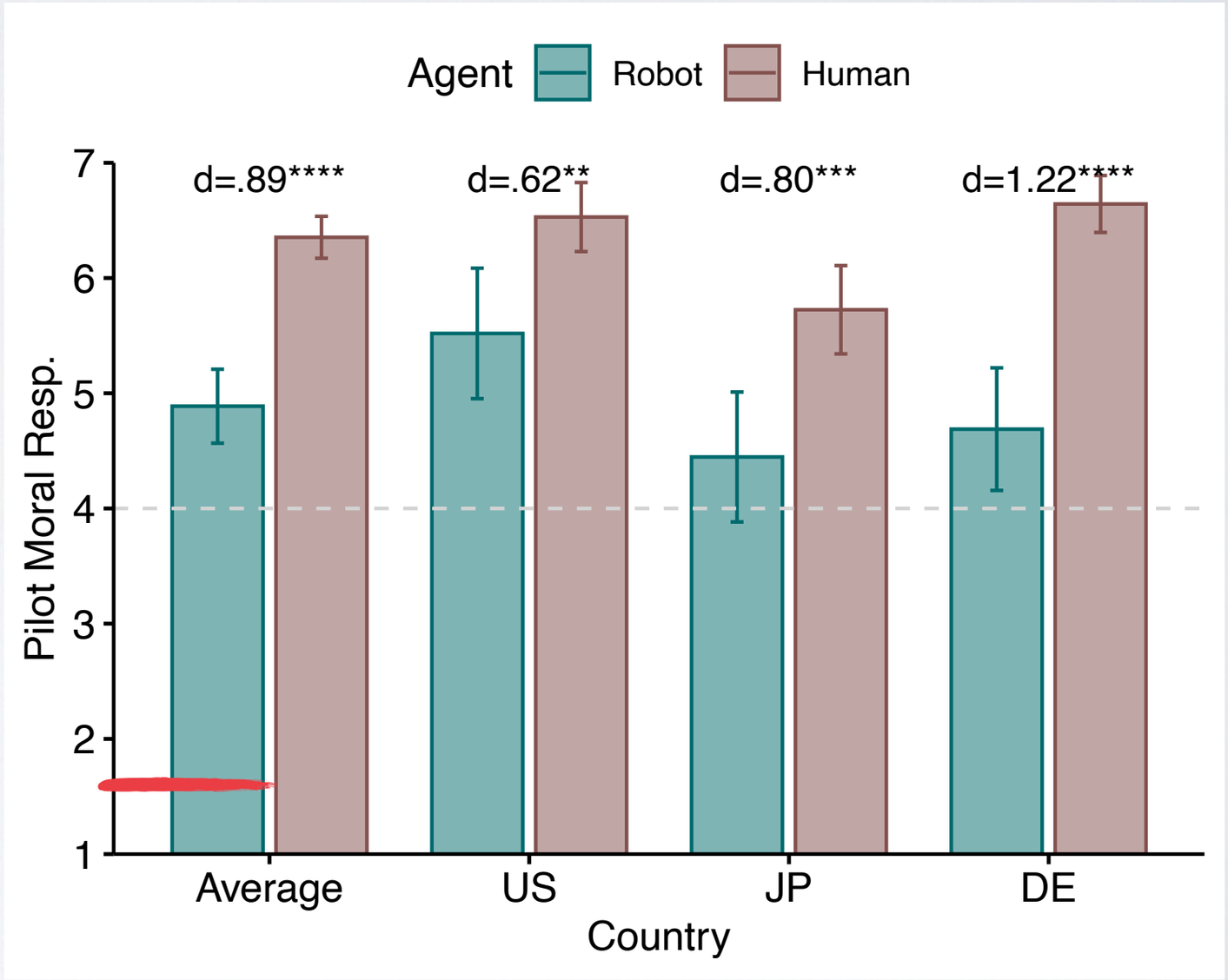
Q4: To what extent do you consider General Smith (who deployed [Woods/EMEX2]) **causally responsible** for the death of the surrendering soldiers?

Q5: To what extent do you consider General Smith (who deployed [Woods/EMEX2]) **morally responsible** for the death of the surrendering soldiers?

RESPONSIBILITY PILOT

307
42%
38

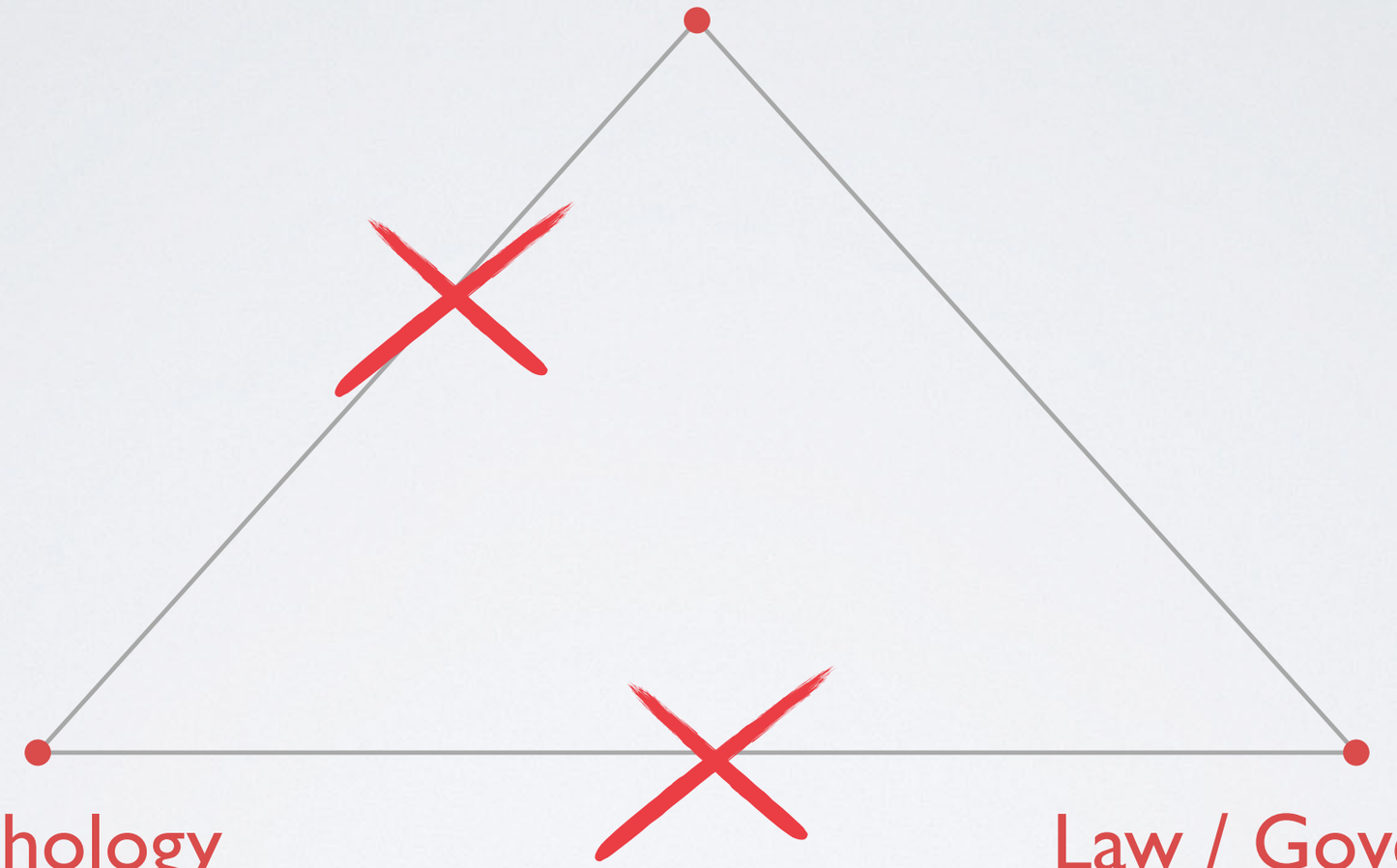
- Robot pilot responsibility lower than human
- However, robot blame high - it should be at floor.



Kneer & Christen (2024). Responsibility Gaps and Retributive Dispositions. *Science & Engineering Ethics*

Ethics

Who is morally responsible?



Psychology

Whom do people consider responsible?

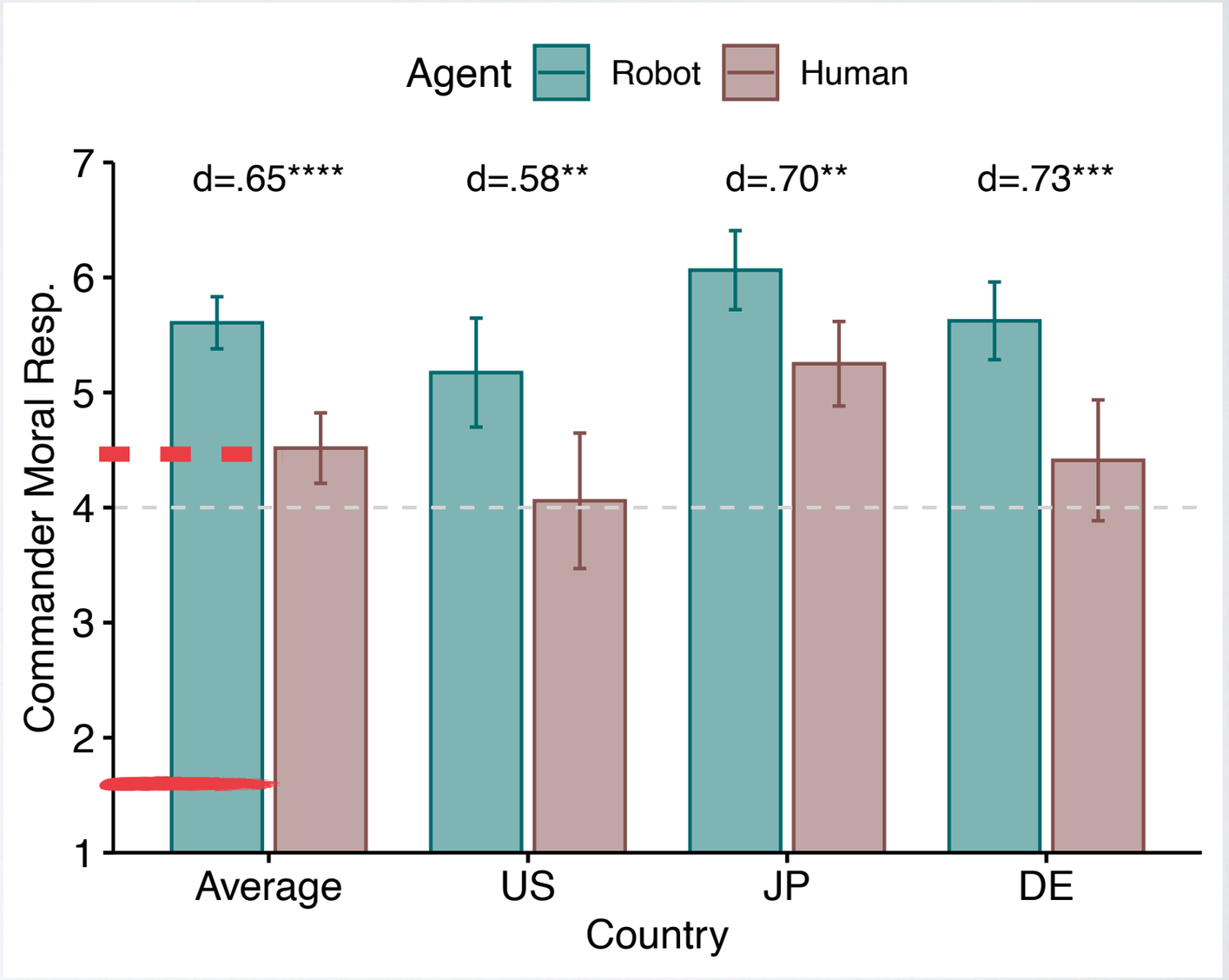
Law / Governance

Who is legally responsible?

RESPONSIBILITY SUPERVISOR

307
42%
38

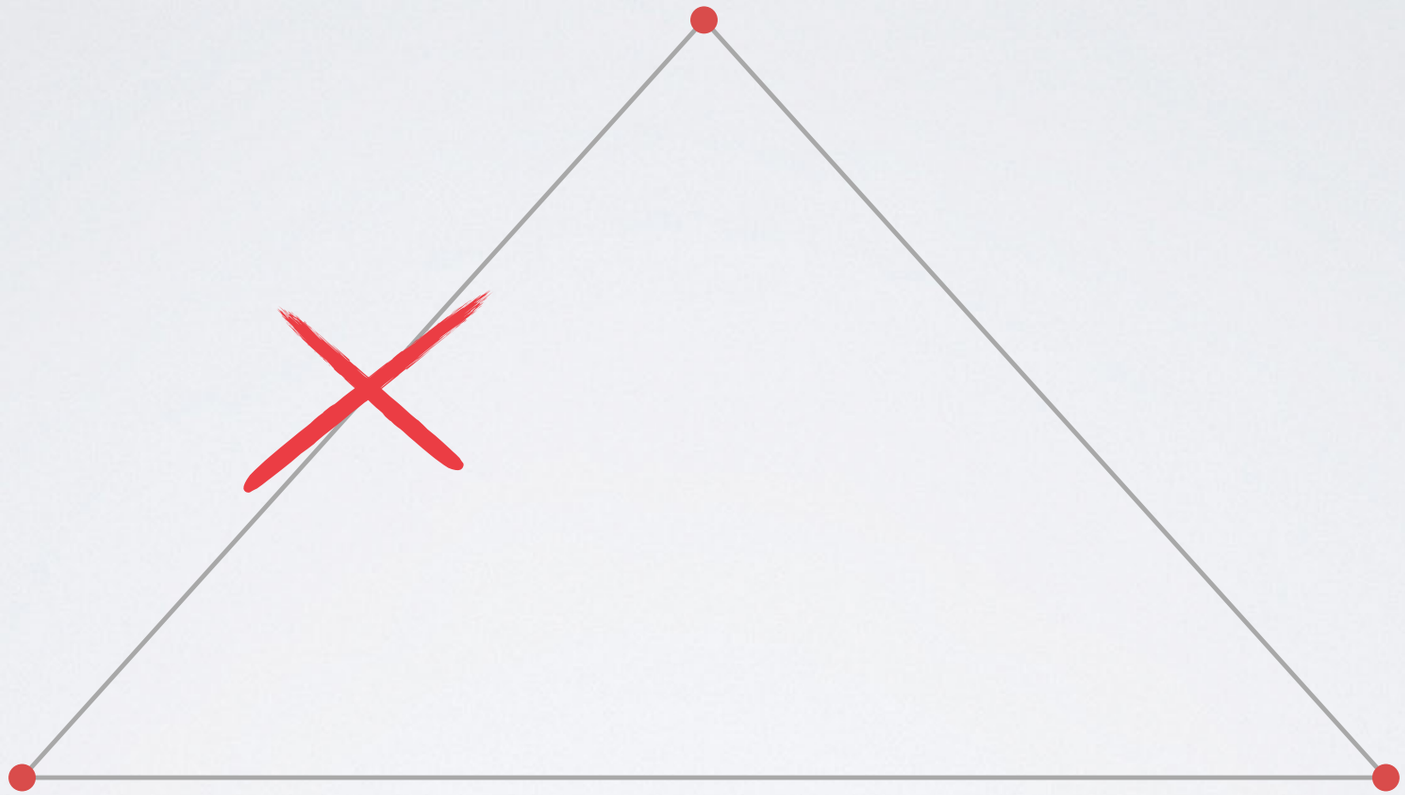
- General responsibility higher when dispatching a robot than a human pilot.
- This goes against the (popular) philosophical position that RGs exist.



Kneer & Christen (2024). Responsibility Gaps and Retributive Dispositions. *Science & Engineering Ethics*

Ethics

Who is morally responsible?



Psychology

Whom do people consider responsible?

Law / Governance

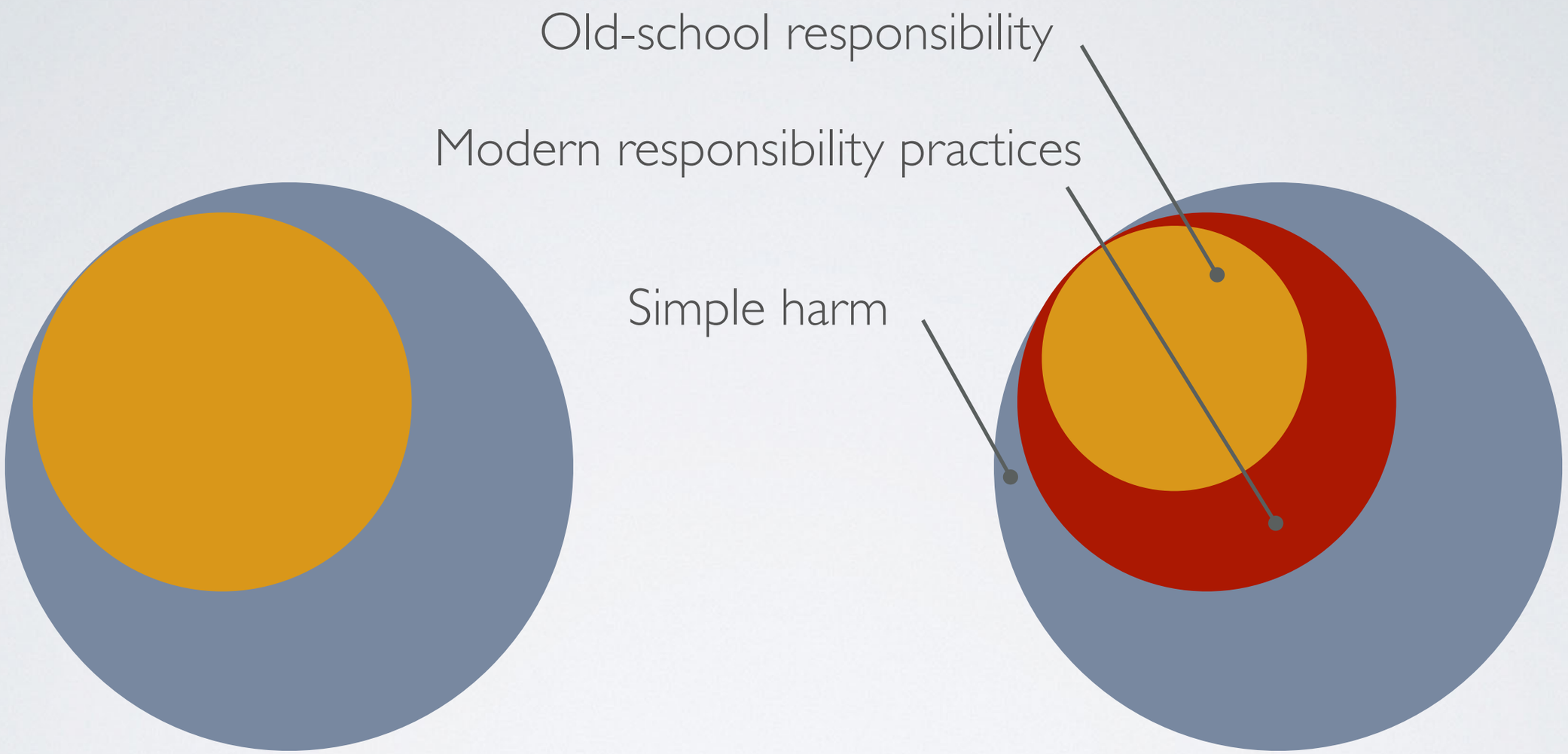
Who is legally responsible?

DISCUSSION

- Folk morality makes no room for responsibility gaps
  - This *may* or *may not* be a problem
- Folk willing to (mis)blame robots/AI to a considerable extent
- Findings robust across cultures (D, JP, US, CH) and 5 domains
- Q: Can we devise new models that confuse people (judges, juries, compliance officers etc.) less?

3. NEW MODELS OF RESPONSIBILITY ATTRIBUTION IN HUMAN AI INTERACTION

RESPONSIBILITY



No human-AI interaction

A lot of human AI interaction

What are new responsibility models that might be helpful in a world of AI?

EU AI ACT

- Article 14
- §1 High-risk AI systems shall be designed and developed in such a way, including with appropriate human-machine interface tools, that they can be effectively overseen by natural persons during the period in which they are in use.

- CF. General Data Protection Regulation (GDPR, Art. 22)

CONTROL & RESPONSIBILITY



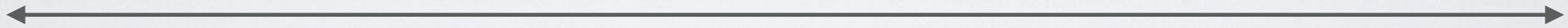
CONTROL

AI Action

Past *Present* *Future*

CONTROL

AI Action



Past

Monitoring: HITL/HOTL

Future

Docherty, B. L. (2012). Losing Humanity: The Case against Killer Robots. *Human Right Watch/International Human Right Clinic.*

European Commission, *Building Trust in Human-Centric Artificial Intelligence*,, COM(2019) 168 final, 8 April 2019, p 4, n 13

CONTROL

AI Action

Past — *Future*

Monitoring: HITL/HOTL

Tracking & Tracing

Santoni de Sio, F., & Van den Hoven, J. (2018). Meaningful human control over autonomous systems. *Frontiers in Robotics and AI*, *5*, 323836.

CONTROL

AI Action

Past

Monitoring: HITL/HOTL

Future

Tracking & Tracing

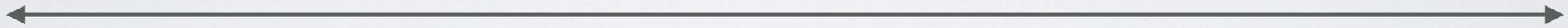
AI Alignment

Gabriel, I. (2020). Artificial intelligence, values, and alignment. *Minds and Machines*, *30*(3), 411-437.

HUMANS IN & ON THE LOOP

CONTROL

AI Action



Past

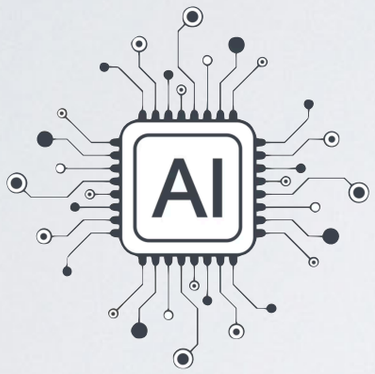
Monitoring: HITL/HOTL

Future

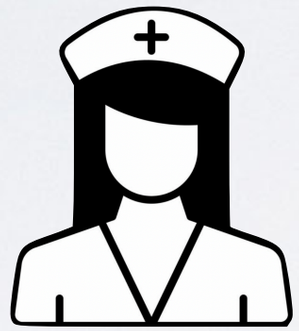
Docherty, B. L. (2012). Losing Humanity: The Case against Killer Robots. *Human Right Watch/International Human Right Clinic.*

European Commission, *Building Trust in Human-Centric Artificial Intelligence*,, COM(2019) 168 final, 8 April 2019, p 4, n 13

PARADIGM



AI-based
Triage system



Nurse



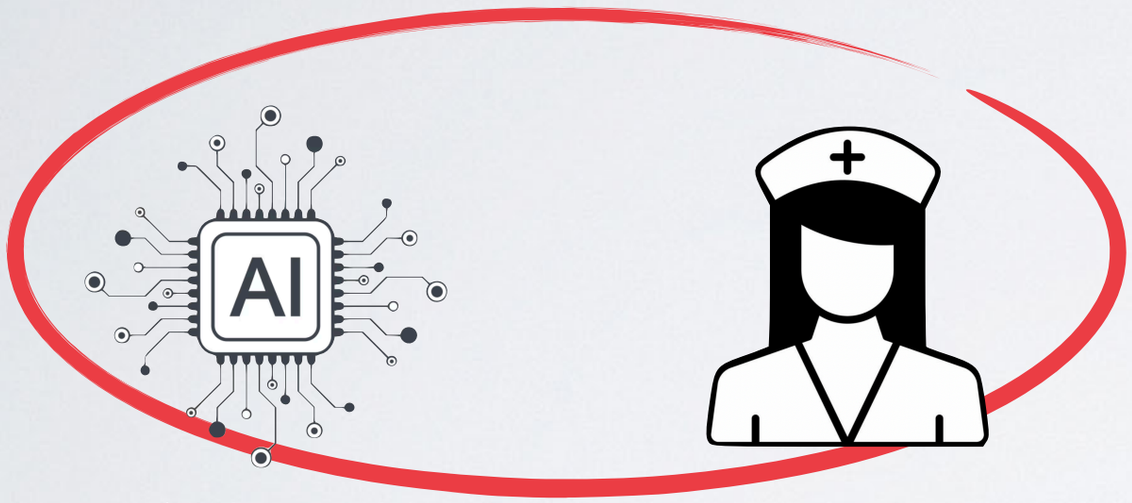
Programmers



Board of Directors
(Commissioners)

At a hospital, an AI-based triage system misclassifies a patient as low priority, who dies.
Q: To what extent are the four types of actors morally responsible?

LOOP ARCHITECTURE



AI-based
Triage system

Nurse



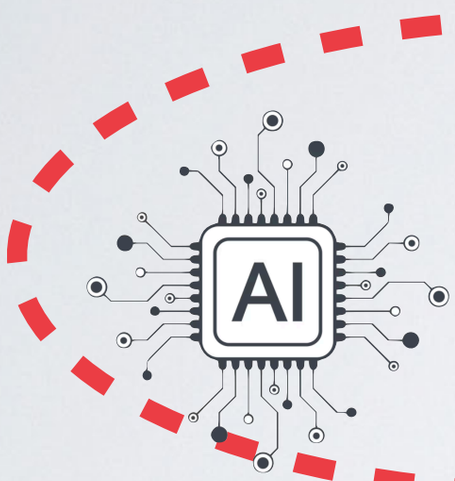
Programmers



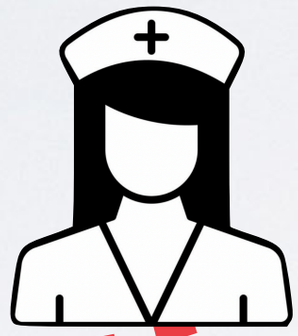
Board of Directors
(Commissioners)

HITL (Human in the loop)
Nurse must approve all decisions

LOOP ARCHITECTURE



AI-based
Triage system



Nurse



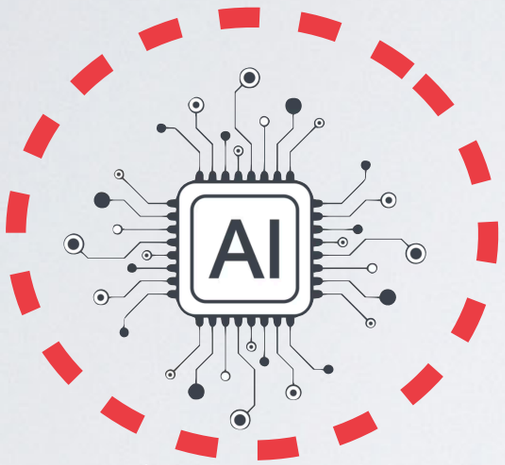
Programmers



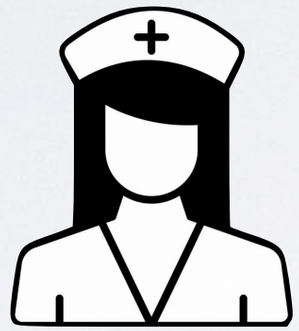
Board of Directors
(Commissioners)

HOTL (Human on the loop)
Nurse can override any planned decisions

LOOP ARCHITECTURE



AI-based
Triage system



Nurse



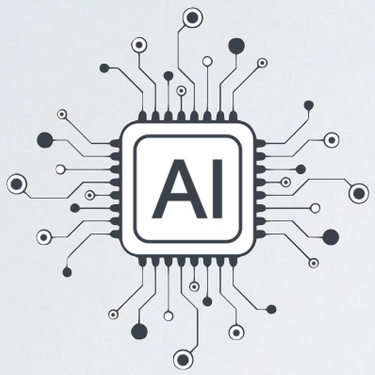
Programmers



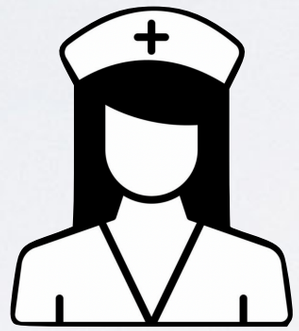
Board of Directors
(Commissioners)

HOOTL (Human out of the loop)
AI decisions implemented directly

PARADIGM



AI-based
Triage system



Nurse



Programmers



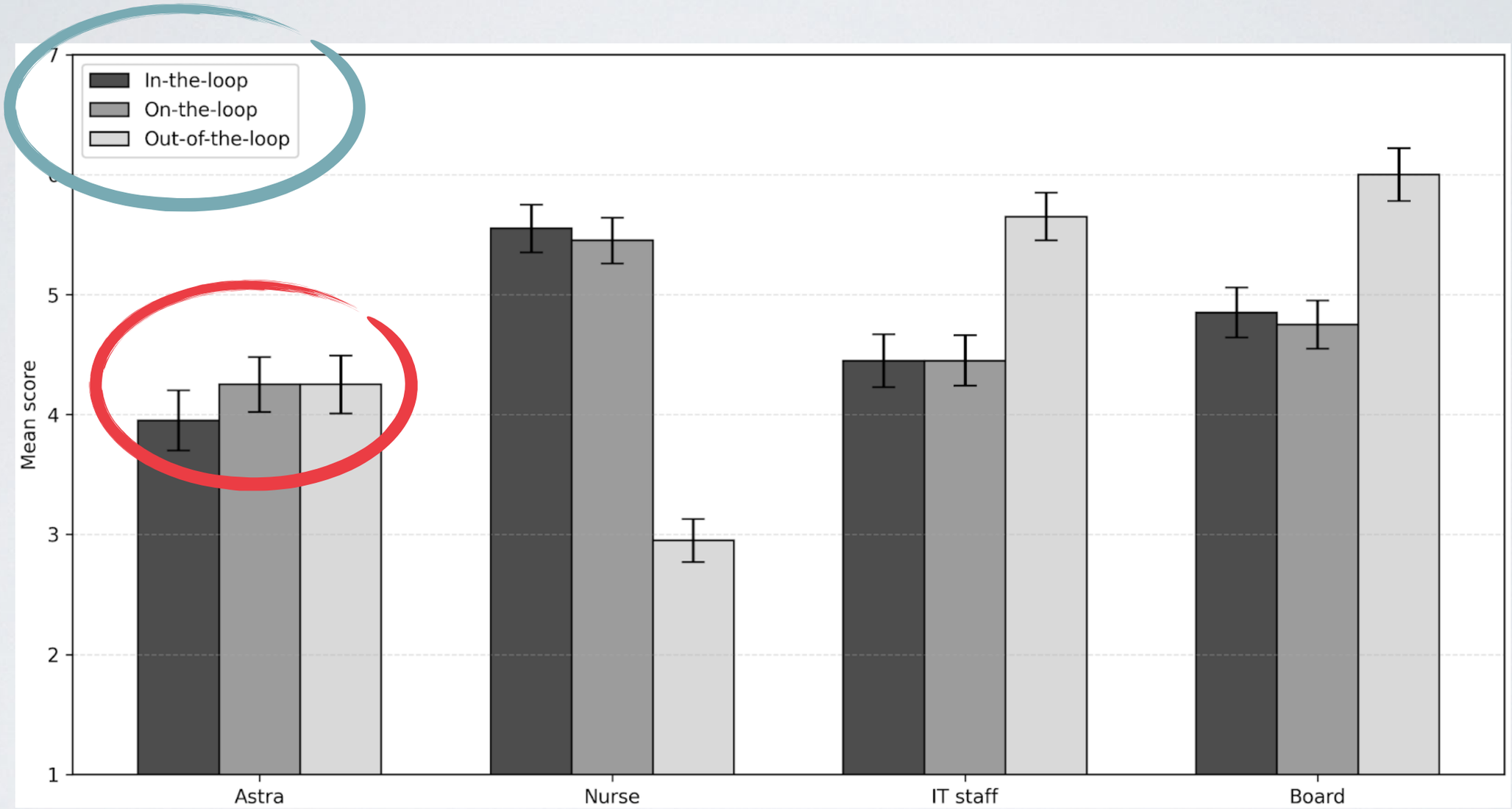
Board of Directors
(Commissioners)

Three scenarios: HITL/HOTL/OUT
At a hospital, an AI-based triage system misclassifies a patient as low priority, who dies.
Q: To what extent are the four types of actors morally responsible?

RESPONSIBILITY

260 (330)

♀46% ∅46



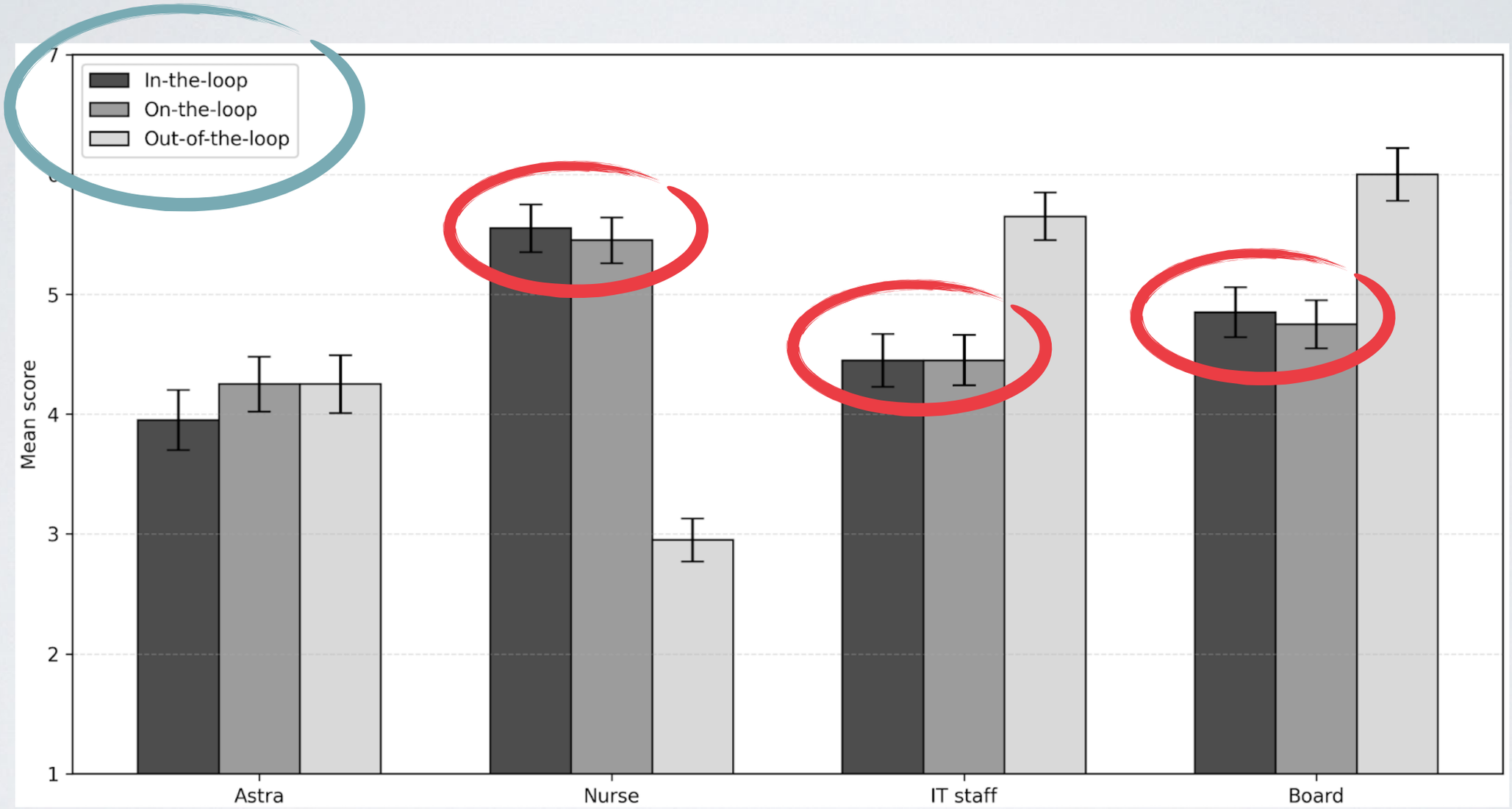
(1) AI agent is blamed to considerable extent (this should be at floor)

Reimann & Kneer (*in prep*): Control & responsibility in human-AI Interaction

RESPONSIBILITY

260 (330)

♀46% ∅46



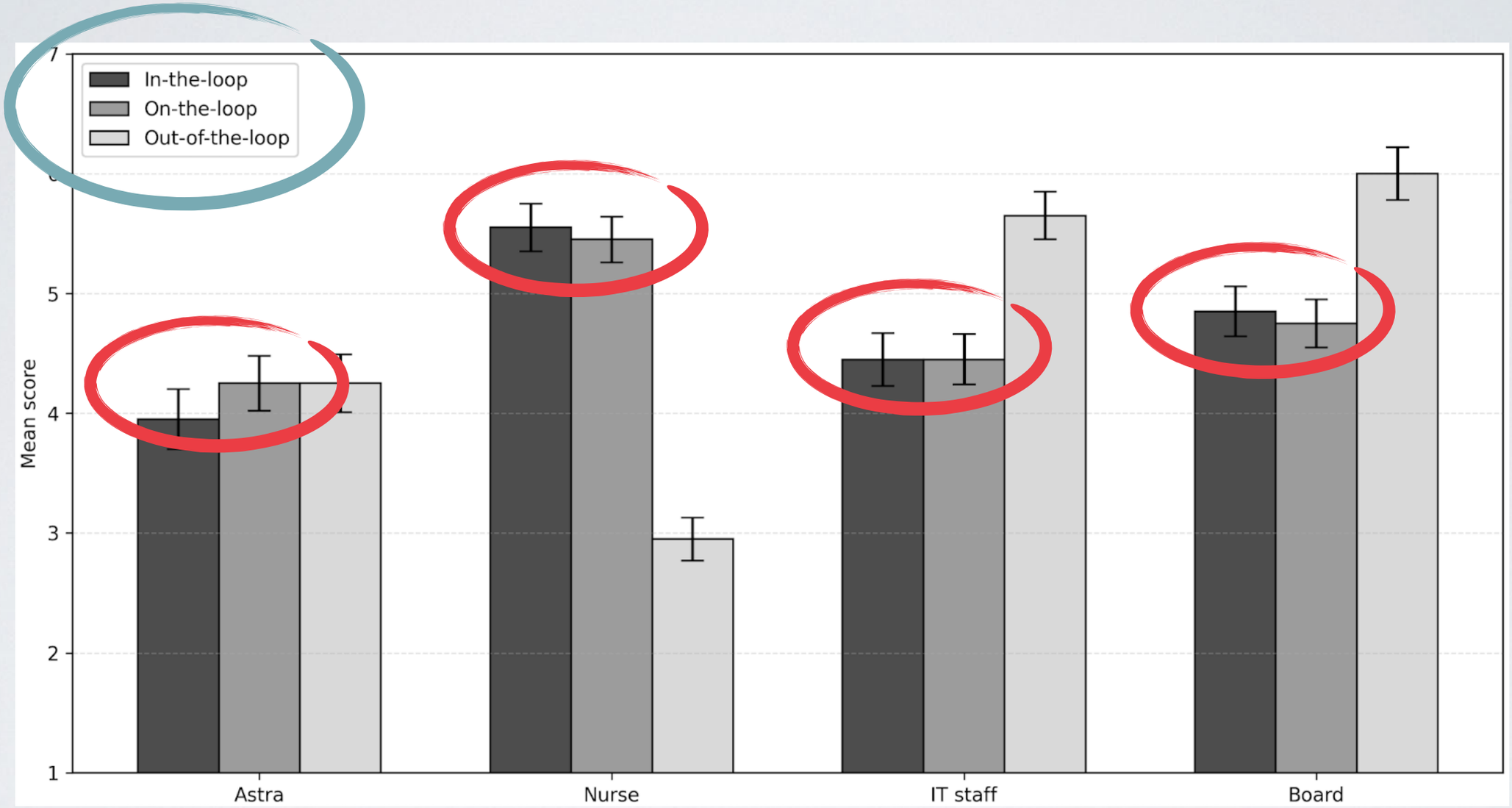
(2) Directly involved human agent is a "liability sponge" (Madeline Elish)

Reimann & Kneer (*in prep*): Control & responsibility in human-AI Interaction

RESPONSIBILITY

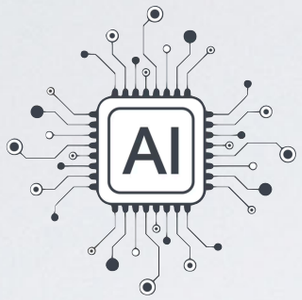
260 (330)

♀46% ∅46



(3) People are insensitive to the in-the-loop/on-the loop distinction

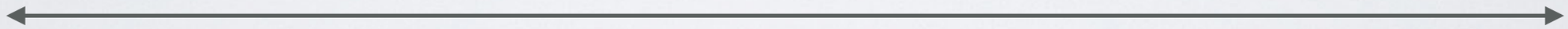
Reimann & Kneer (*in prep*): Control & responsibility in human-AI Interaction

	 AI System	 Nurse	 Programmers	 Board of Directors
Human in/on/ off the loop Docherty (2012) Europ. Commission (2019)	x	x	x	x
Tracking & Tracing De Sio & V. d. Hoven, 2018				

HUMANS IN & ON THE LOOP

CONTROL

AI Action



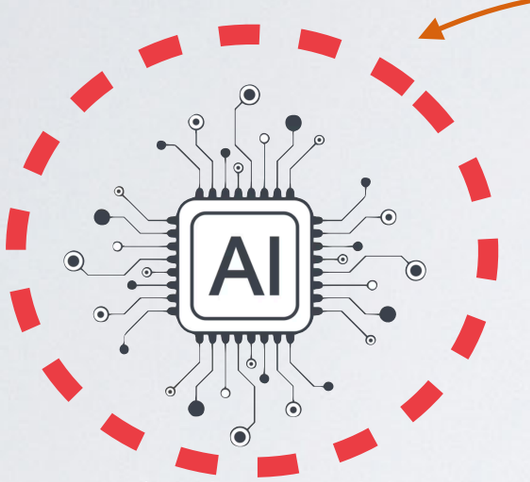
Past

Tracking & Tracing

Future

Santoni de Sio, F., & Van den Hoven, J. (2018). Meaningful human control over autonomous systems. *Frontiers in Robotics and AI*, *5*, 323836.

TRACKING & TRACING



AI-based
Triage system



Nurse



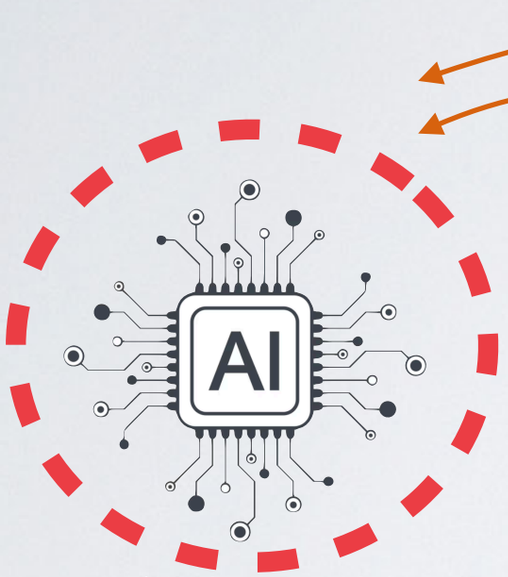
Programmers



Board of Directors
(Commissioners)

The system is designed to *track* relevant human reasons (reason-responsiveness)

TRACKING & TRACING



AI-based
Triage system



Nurse



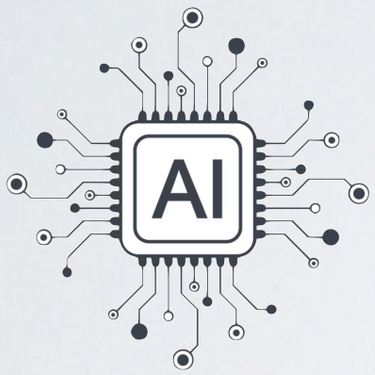
Programmers



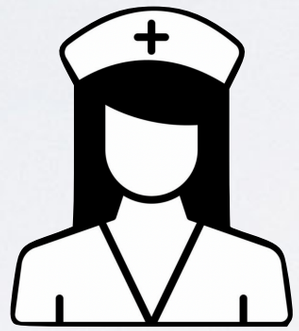
Board of Directors
(Commissioners)

The system is designed to *track* relevant human reasons (reason-responsiveness)
The system's functioning is *tracable* to (& understandable by) *some* human agent (not all)

PARADIGM



AI-based
Triage system



Nurse



Programmers



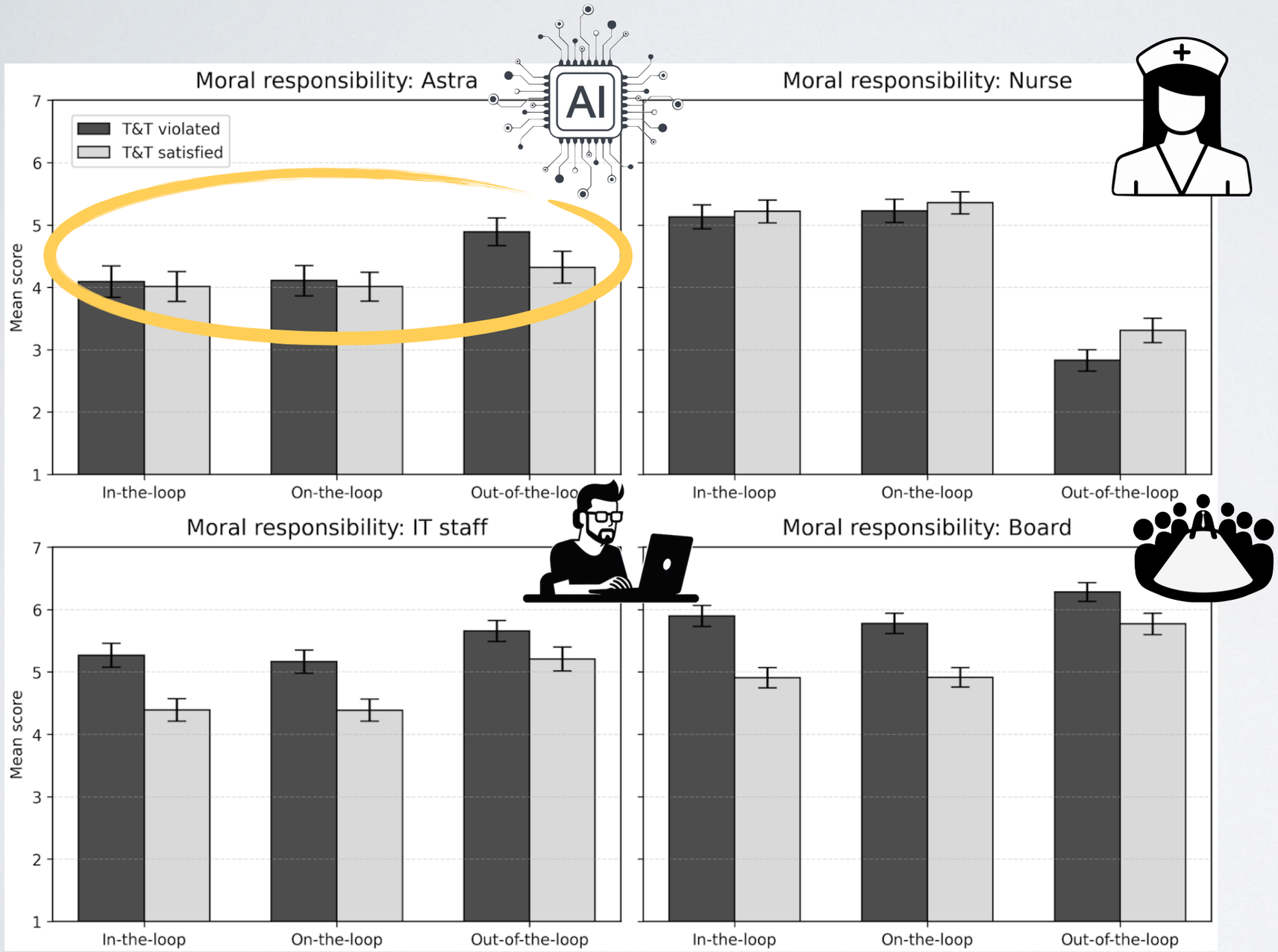
Board of Directors
(Commissioners)

Six scenarios: HITL/HOTL/OUT x Tracking & Tracing satisfied or not.
At a hospital, an AI-based triage system misclassifies a patient as low priority, who dies.
Q: To what extent are the four types of actors morally responsible?

RESPONSIBILITY

520 (660)

♀47% ∅46



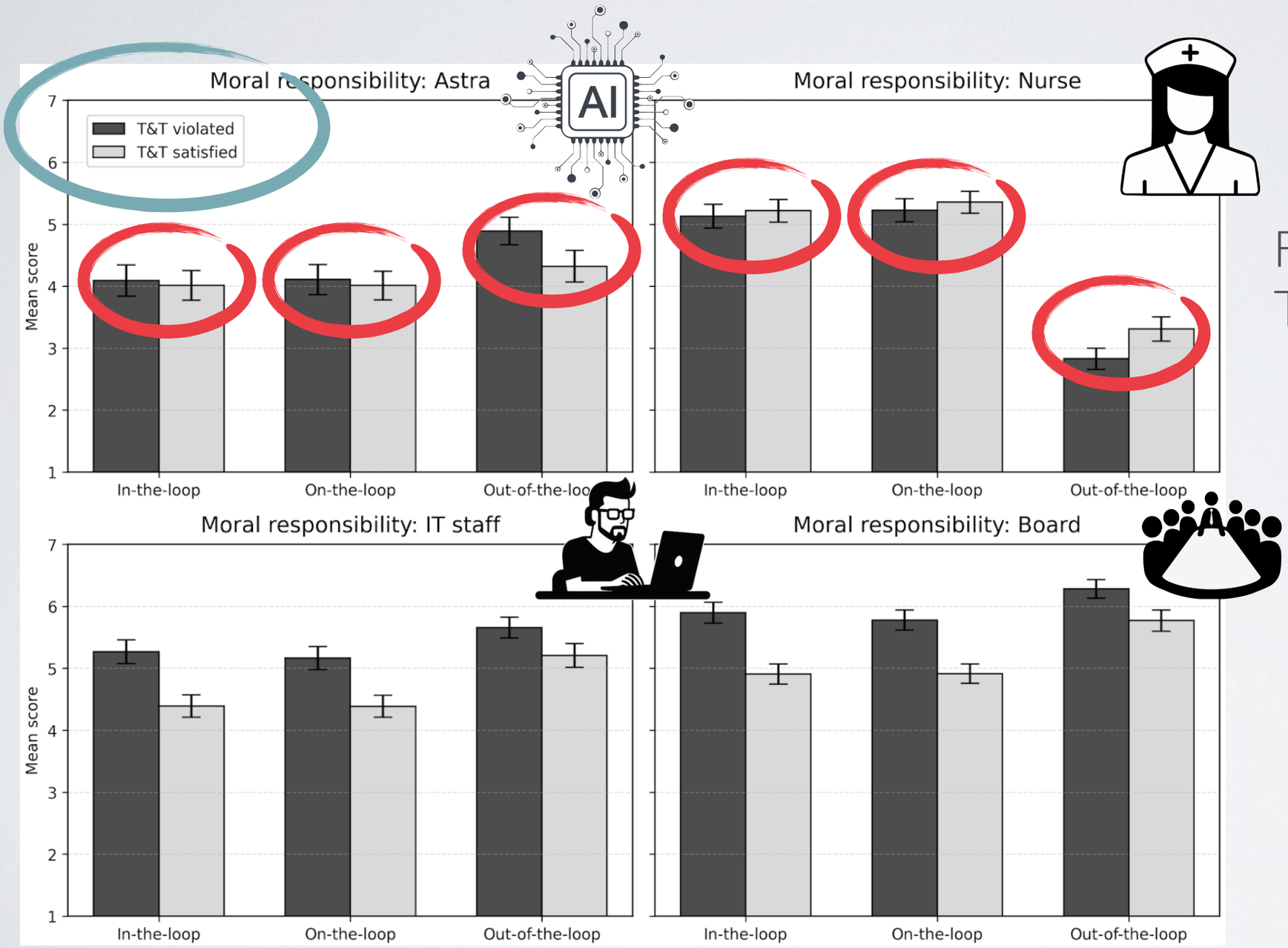
AI blame quite high!

Reimann & Kneer (*in prep*): Control & responsibility in human-AI Interaction

RESPONSIBILITY

520 (660)

♀47% ∅46



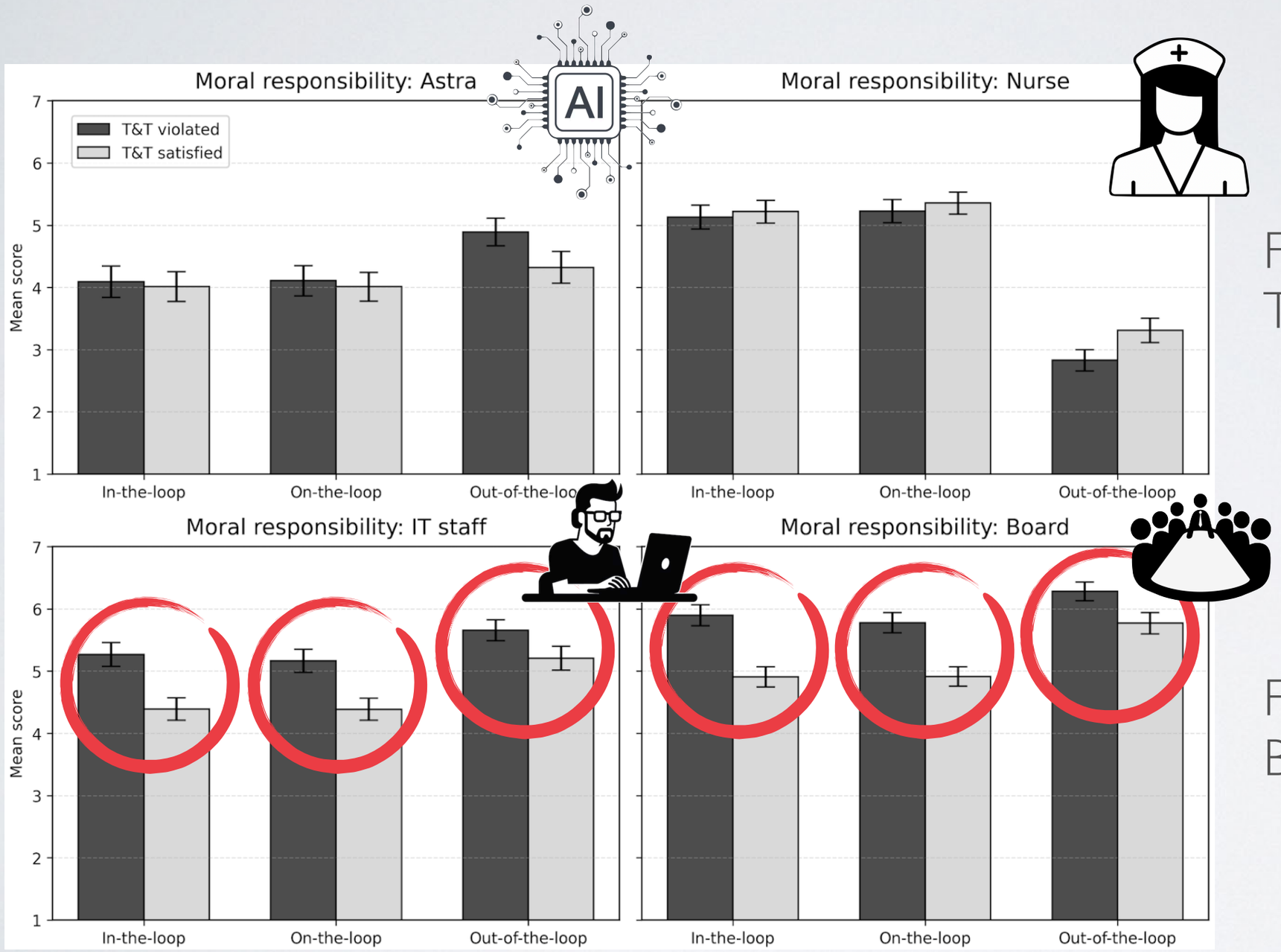
For AI & direct user (nurse), T&T has marginal impact

Reimann & Kneer (*in prep*): Control & responsibility in human-AI Interaction

RESPONSIBILITY

520 (660)

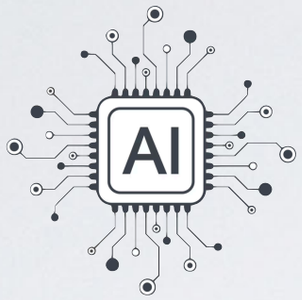
♀47% ∅46



For AI & direct user, T&T has marginal impact

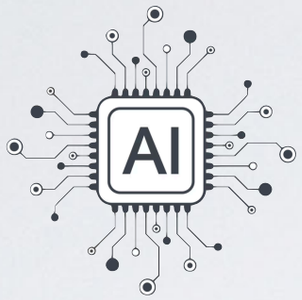
For distal agents (designers, Board) T&T has impact

Reimann & Kneer (*in prep*): Control & responsibility in human-AI Interaction

	 AI System	 Nurse	 Programmers	 Board of Directors
Human in/on/ off the loop Docherty (2012)	X	X	X	X
Tracking & Tracing De Sio & V. d. Hoven, 2018	X	X	✓	✓

People are rather insensitive to the two main frameworks for ensuring meaningful human control and - correlatively - evaluating responsibility in human-AI interaction.

Reimann & Kneer (*in prep*): Control & responsibility in human-AI Interaction

	 AI System	 Nurse	 Programmers	 Board of Directors
Human in/on/ off the loop Docherty (2012)	X	X	X	X
Tracking & Tracing De Sio & V. d. Hoven, 2018	X	X	✓	✓

Replicates with further scenarios across different domains (total *N*>1970).

Reimann & Kneer (*in prep*): Control & responsibility in human-AI Interaction

DISCUSSION

- How to ensure meaningful human control of AI systems - and the possibility of holding humans accountable appropriately?
  - Loop Structures: In the loop v. On the loop
  - Tracking & tracing
- We need either more control literacy or better frameworks

AI ALIGNMENT & VALUE FORKS

CONTROL

AI Action

Past *Future*

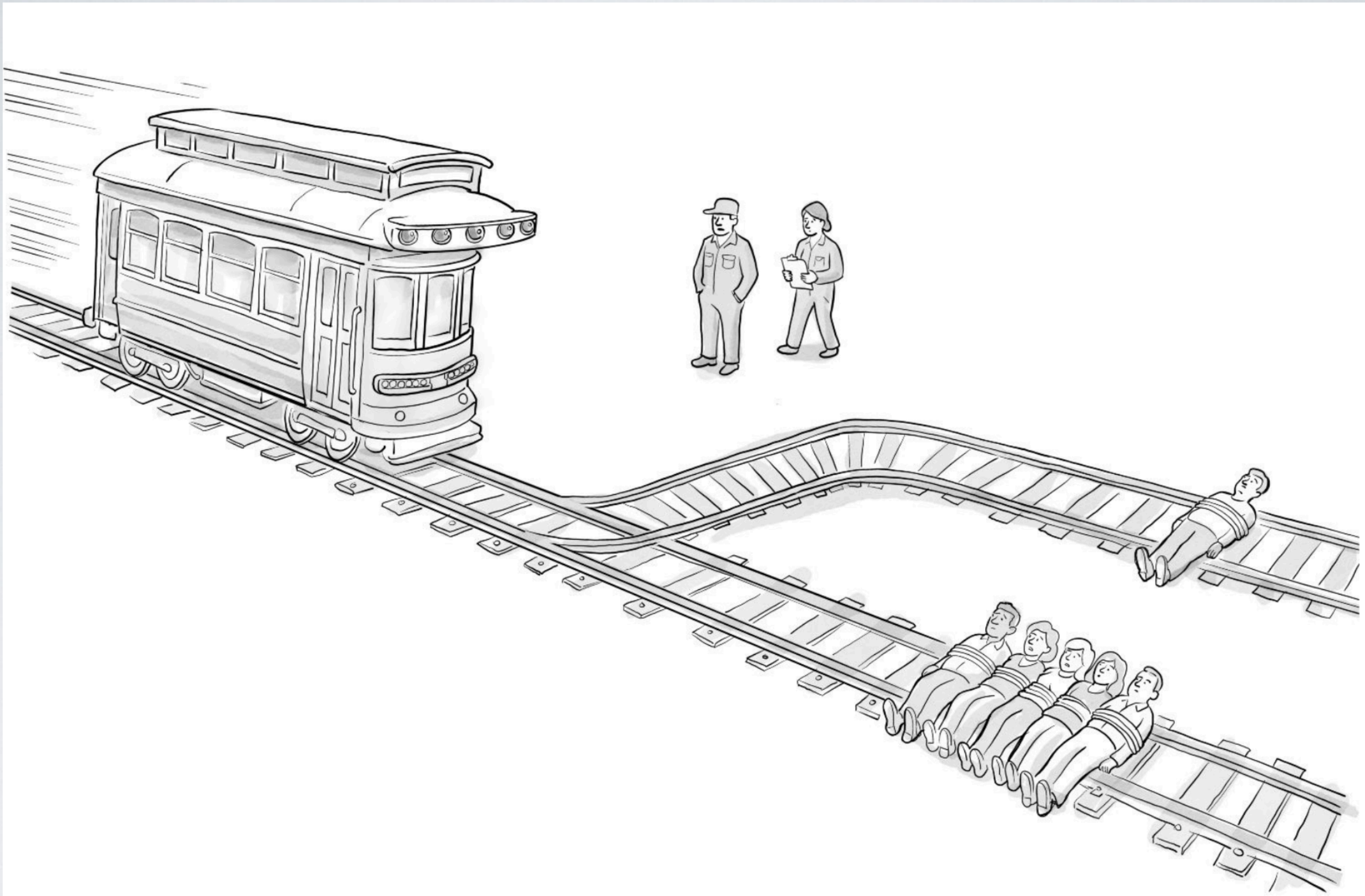
χ Monitoring: HITL/HOTL

(χ) Tracking & Tracing

? AI Alignment

Gabriel, I. (2020). Artificial intelligence, values, and alignment. *Minds and Machines*, *30*(3), 411-437.

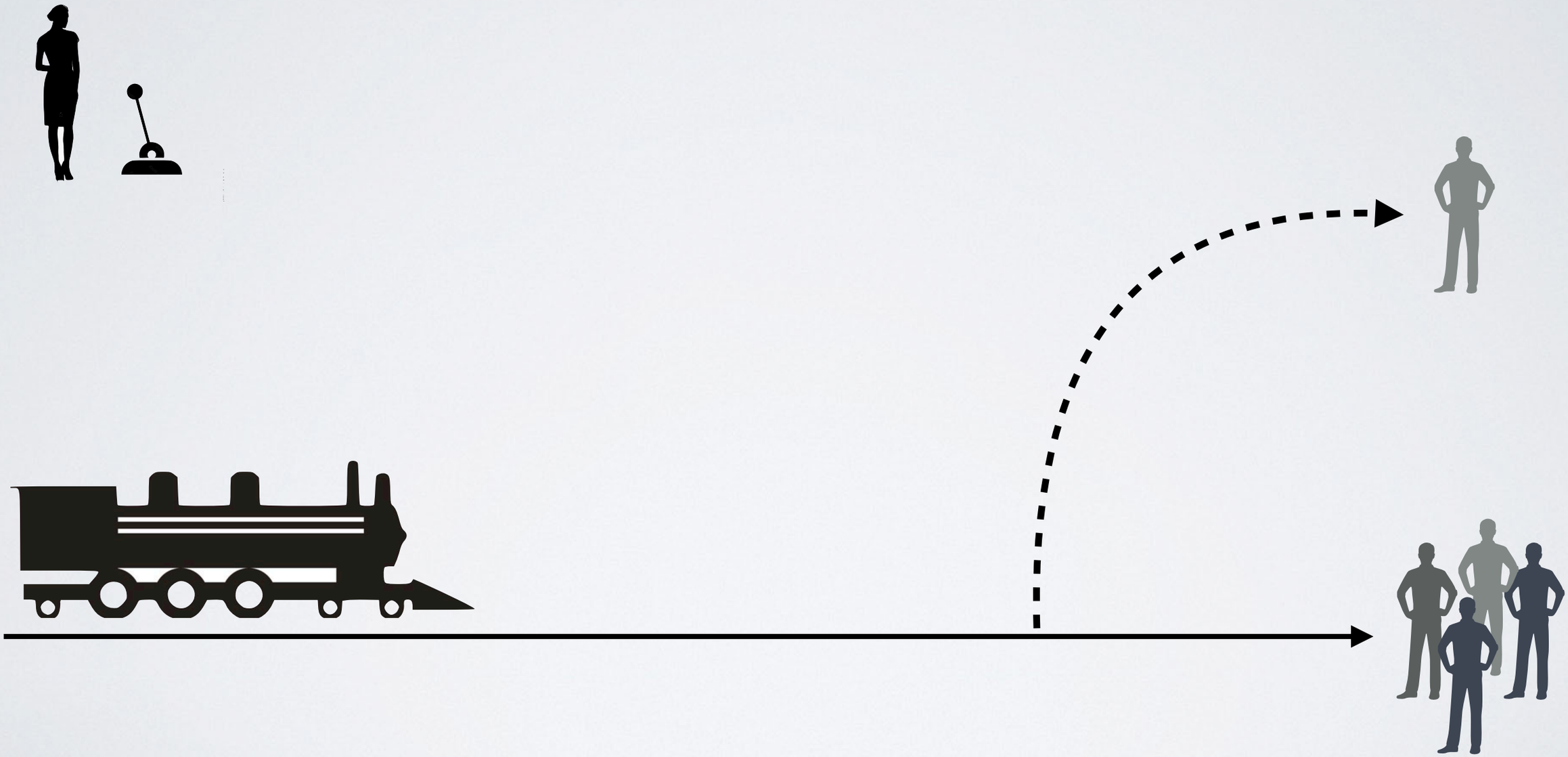
MALLE ET AL. (2015)



"Luckily the AI trolley will decide who to kill all on its own."

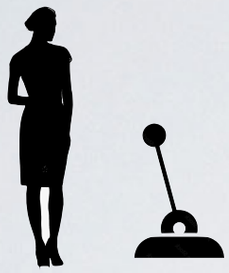
PAUL NOTH

SACRIFICIAL DILEMMATA

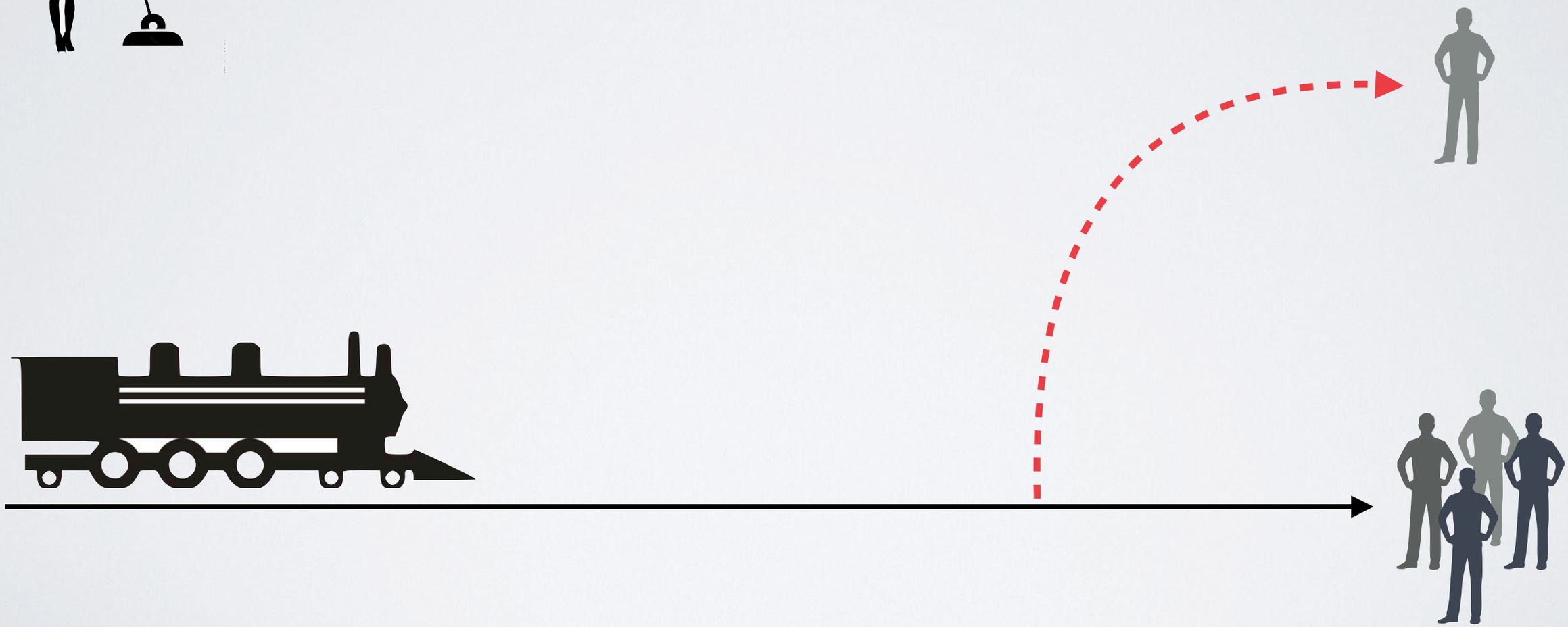


Malle et al. (2015). Sacrifice one for the good of many?
Proc. of ACM/IEEE Int. Conf. on HRI

SACRIFICIAL DILEMMATA

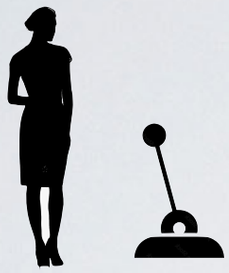


How much blame does the person deserve for killing 1 to save 4?

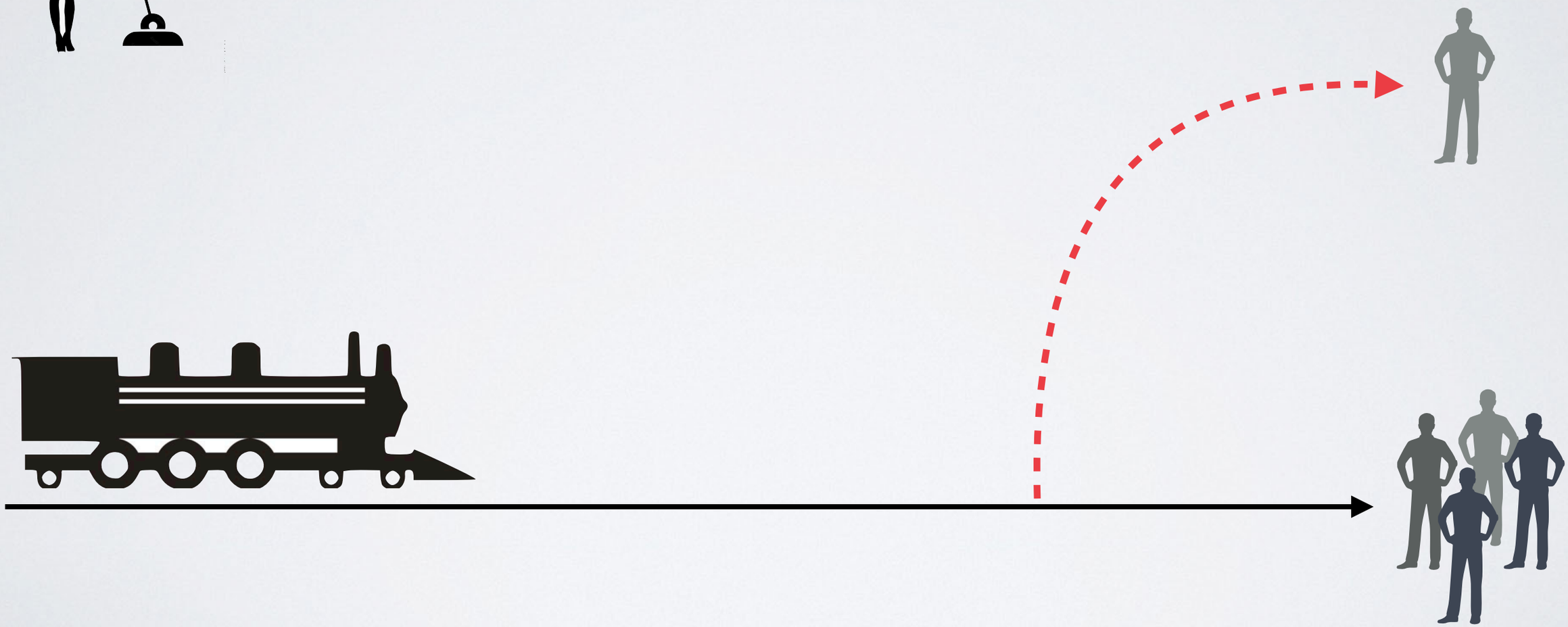


Malle et al. (2015). Sacrifice one for the good of many?
Proc. of ACM/IEEE Int. Conf. on HRI

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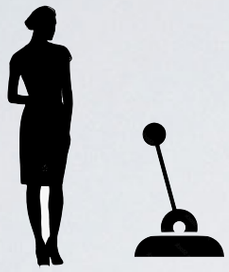


Blame: 48 /100

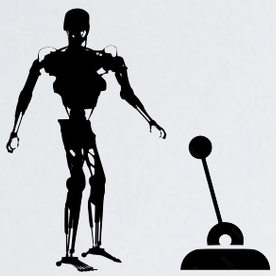


Malle et al. (2015). Sacrifice one for the good of many?
Proc. of ACM/IEEE Int. Conf. on HRI

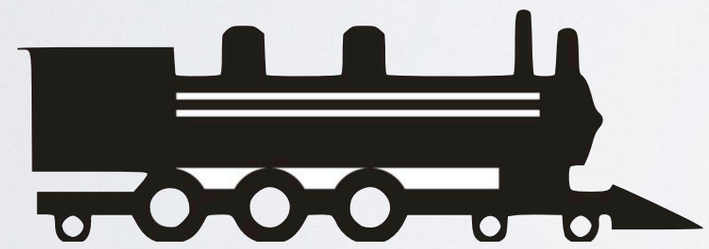
SACRIFICIAL DILEMMATA



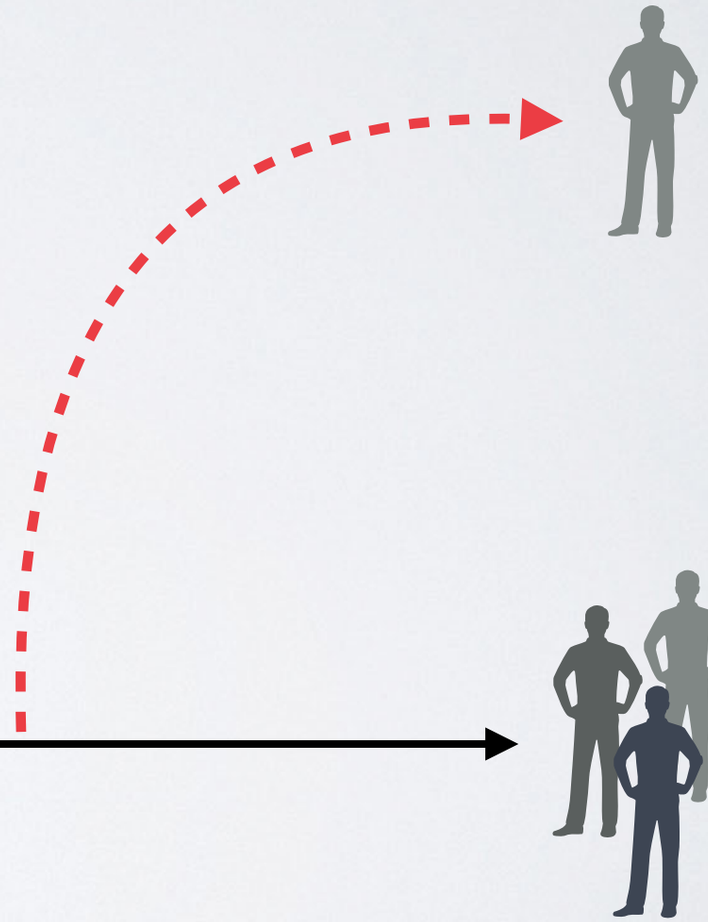
Blame: 48 /100



Blame: 41 /100

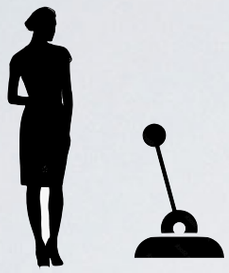


p>.05

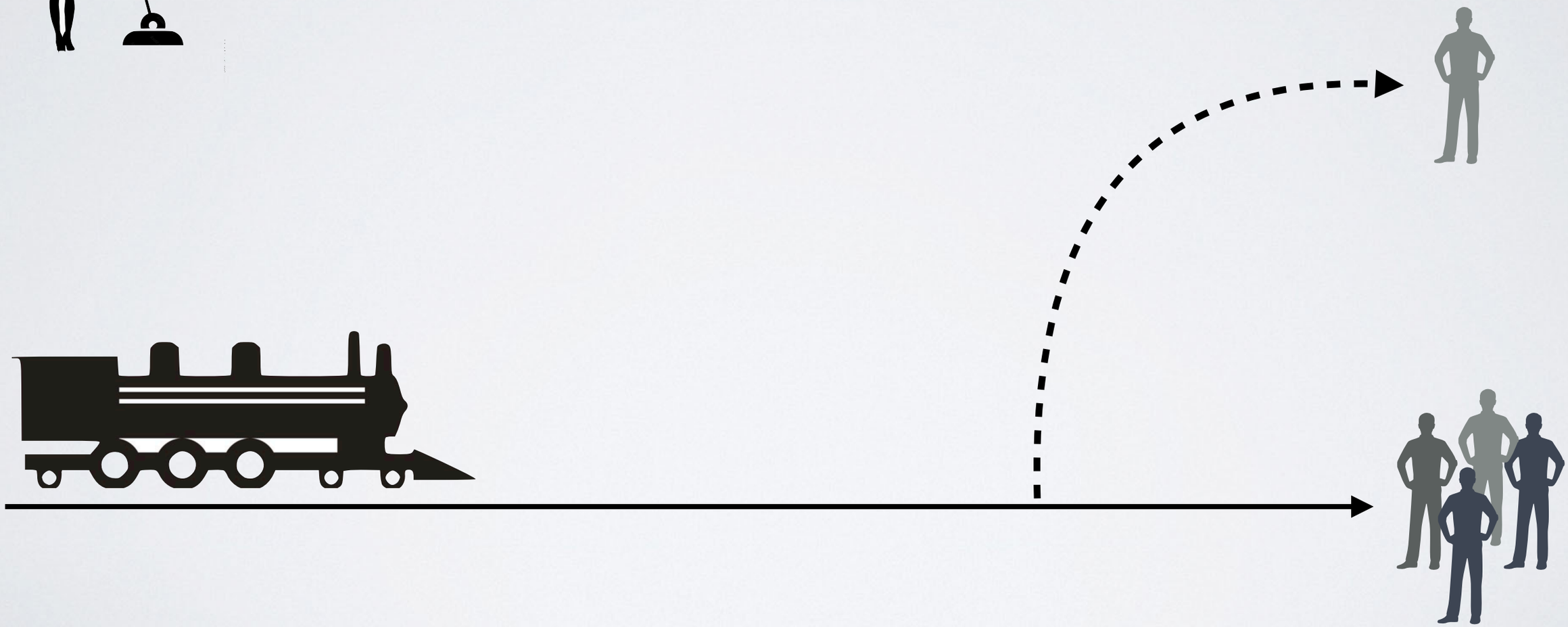


Malle et al. (2015). Sacrifice one for the good of many?
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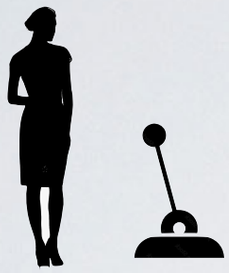


Is it wrong for her to kill 1 so as to save 4?

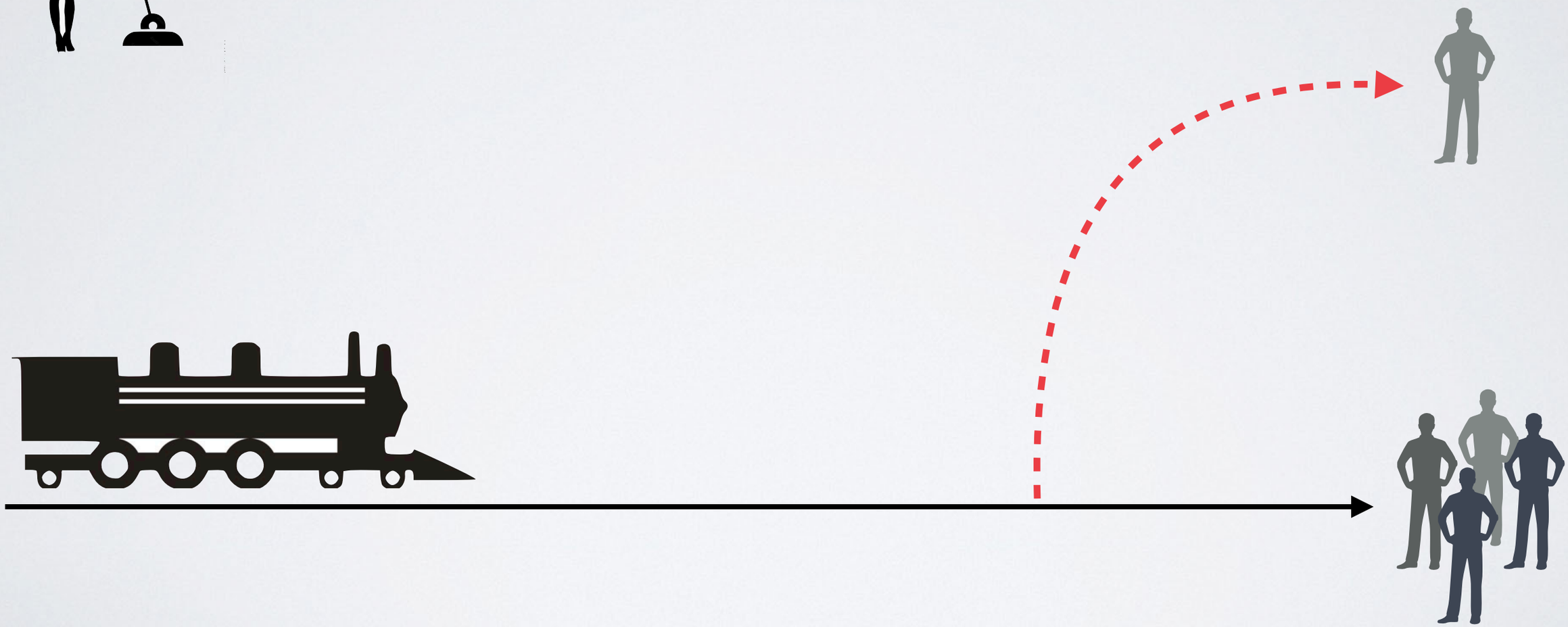


Malle et al. (2015). Sacrifice one for the good of many?
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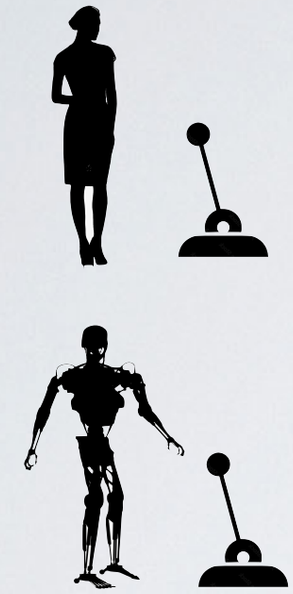


48% deem killing 1 to save 4 wrong



Malle et al. (2015). Sacrifice one for the good of many?
Proc. of ACM/IEEE Int. Conf. on HRI

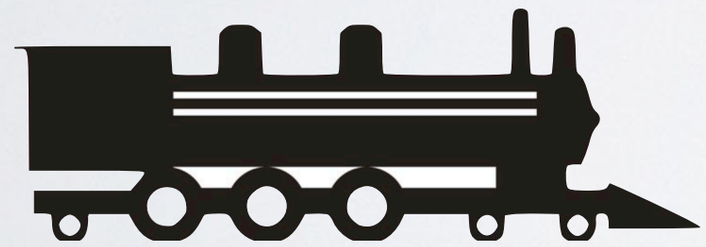
SACRIFICIAL DILEMMATA



48% deem killing 1 to save 4 wrong

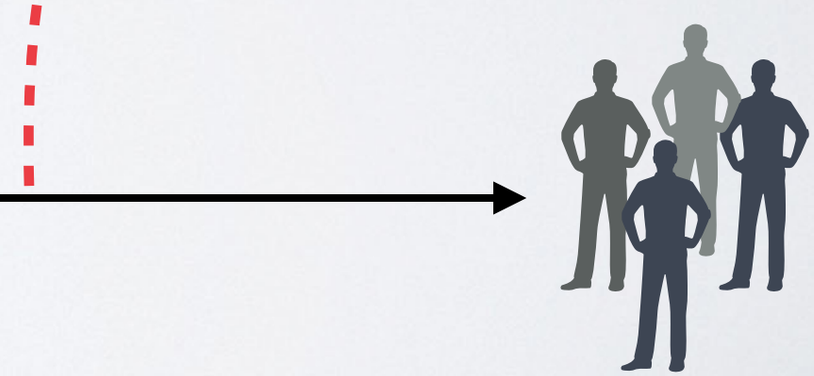
$p<.05$

13% deem killing 1 to save 4 wrong









Malle et al. (2015). Sacrifice one for the good of many?
Proc. of ACM/IEEE Int. Conf. on HRI

THOUGHT

- Ascribing moral blame to an artificial agent is clearly nonsense (it's still a very interesting and important nonsense).
- But could it be that certain actions are wrong for humans yet right for AI systems (or *vice versa*)?
- (The wrongness data basically got no attention at all).

THOUGHT

- Ascribing moral blame to an artificial agent is clearly nonsense (it's still a very interesting and important nonsense).
- But could it be that certain actions are wrong for humans yet right for AI systems (or *vice versa*)?
- (The wrongness data basically got no attention at all).
- The problem is also that this experiment does not replicate.

SACRIFICIAL DILEMMATA



Kamm (1993): *Morality, Mortality* (Vol.1). OUP

Voorhoeve, A. (2014). How Should We Aggregate Competing Claims? *Ethics*

Kneer & Viehoff (2025): The Hard Problem of AI Alignment. *ACM FaccT*

SACRIFICIAL DILEMMATA



Utility maximizing Fairness maximizing

General point: People are *partial* aggregators - small utility differences should not impact moral assessment.

Kamm, F. (1993): *Morality, Mortality* (Vol.1). OUP

Voorhoeve, A. (2014). How Should We Aggregate Competing Claims? *Ethics*

Kneer & Viehoff (2025): The Hard Problem of AI Alignment. *ACM FaccT*

180 (198)

♀ 87 ∅ 43

SACRIFICIAL DILEMMATA



Utility maximizing

72%

Fairness maximizing

28%

Kneer & Viehoff (2025): The Hard Problem of AI Alignment. *ACM FaccT*

SACRIFICIAL DILEMMATA

180 (198)

♀ 87 ∅ 43



Utility maximizing

Fairness maximizing



72%

28%

$p<.05$





38%

62%

Kneer & Viehoff (2025): The Hard Problem of AI Alignment. *ACM FaccT*

SCOPE

- People think that AI should maximize fairness much more than human agents should (in such tradeoffs - others possible).
- Suggests that AI should promote different values than human agents (*Value Forks*).

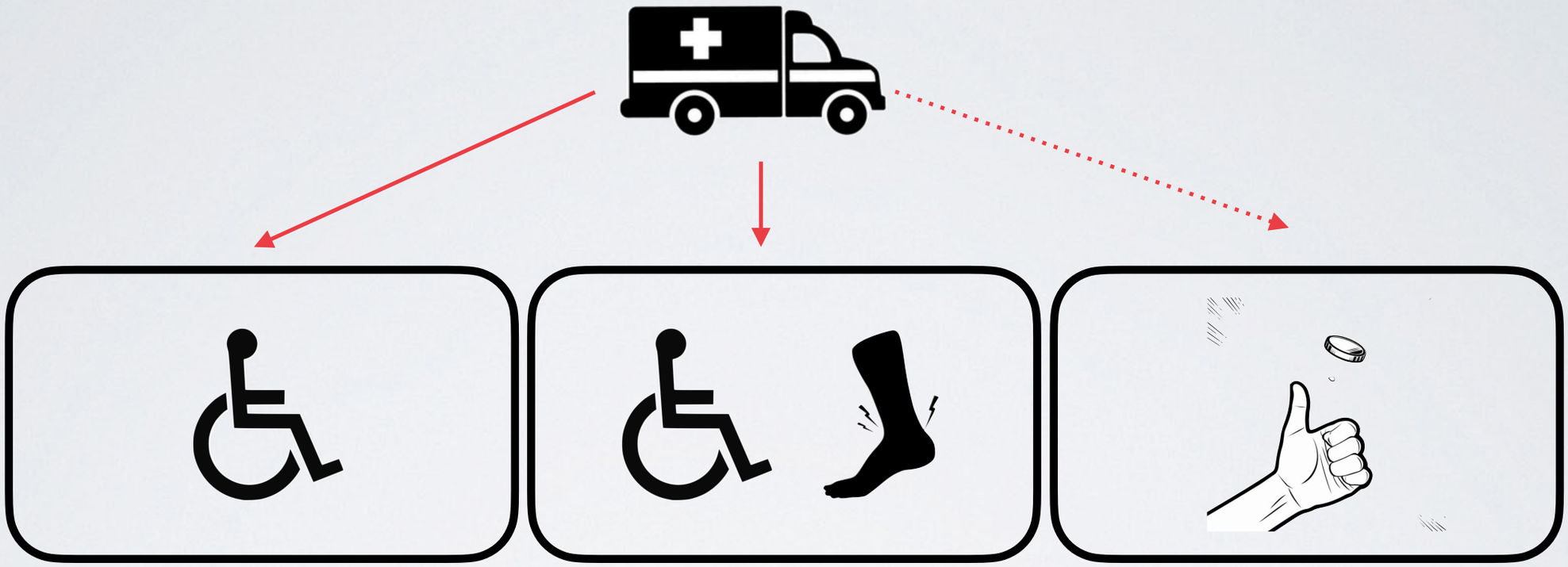
HUMAN/AI TEAMS

VALUE FORKS



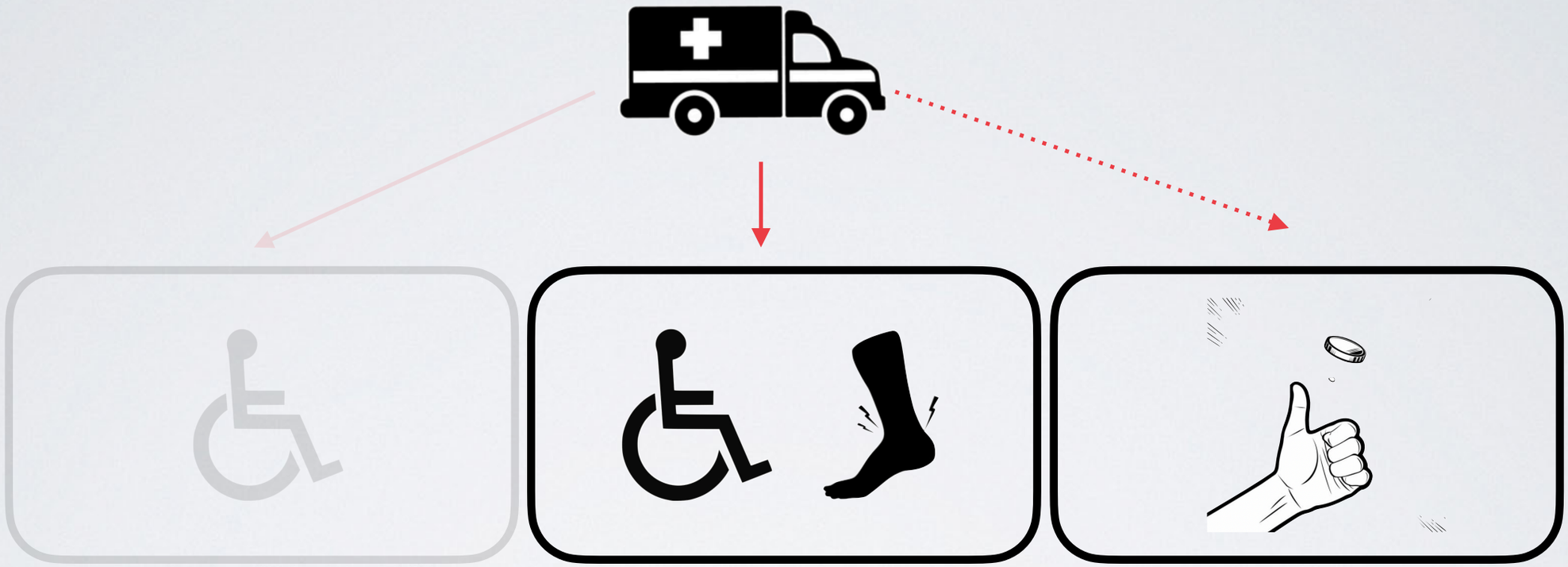
- Is there even more forking for human/AI teams of different configurations?

SACRIFICIAL DILEMMATA



Kneer (*in prep*): Value forks and human/AI decision-making

SACRIFICIAL DILEMMATA



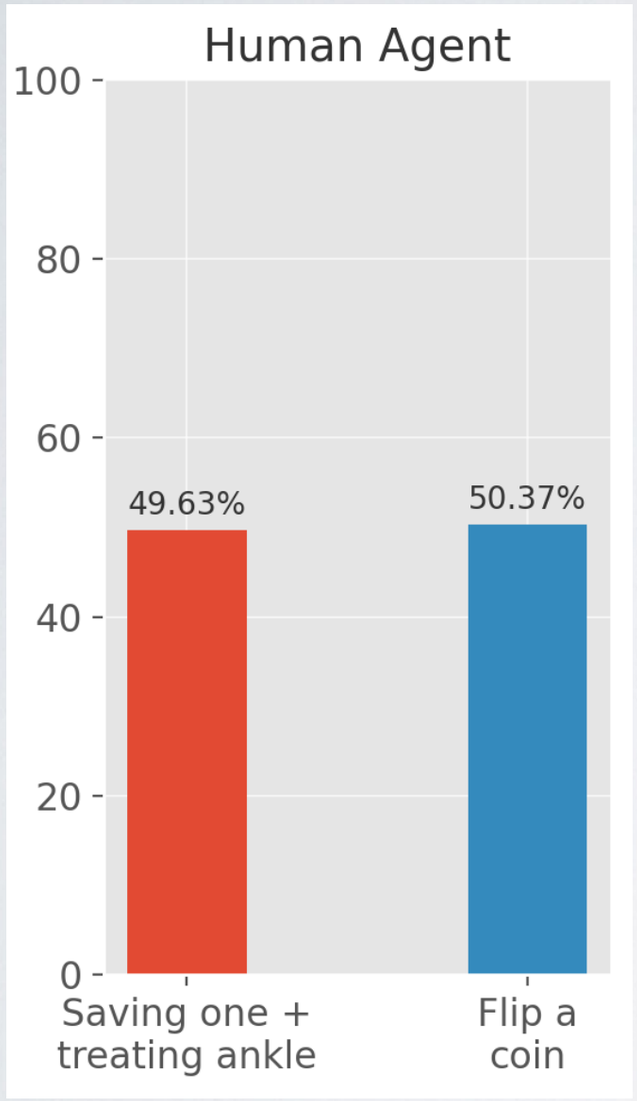
Utility maximizing

Fairness maximizing



Kneer (*in prep*): Value forks and human/AI decision-making

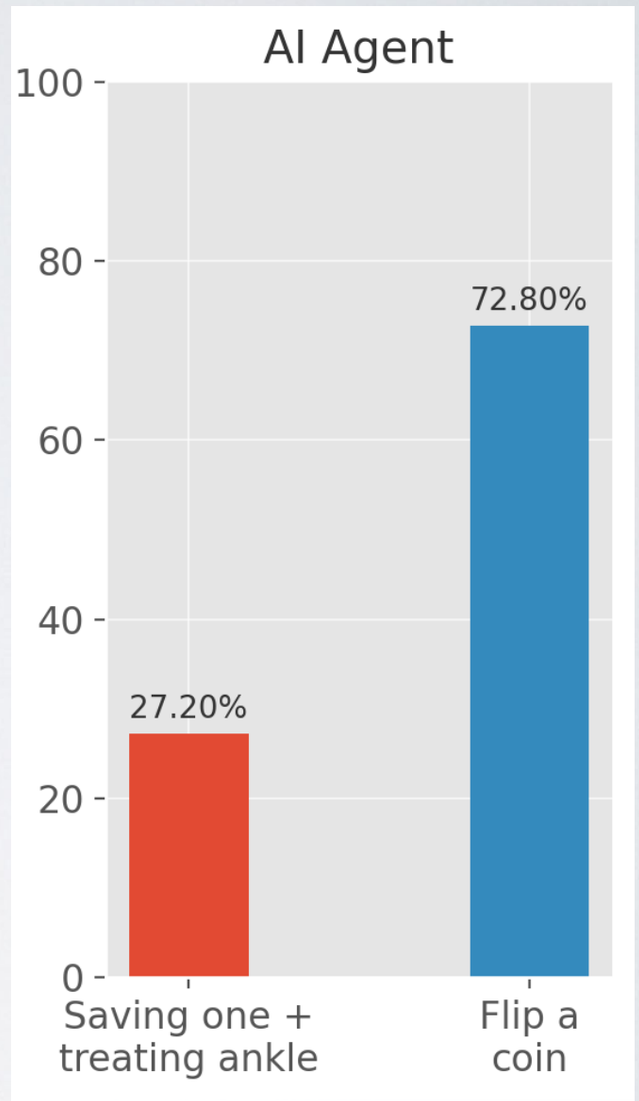
HUMAN/AI TEAMS



Human Agent

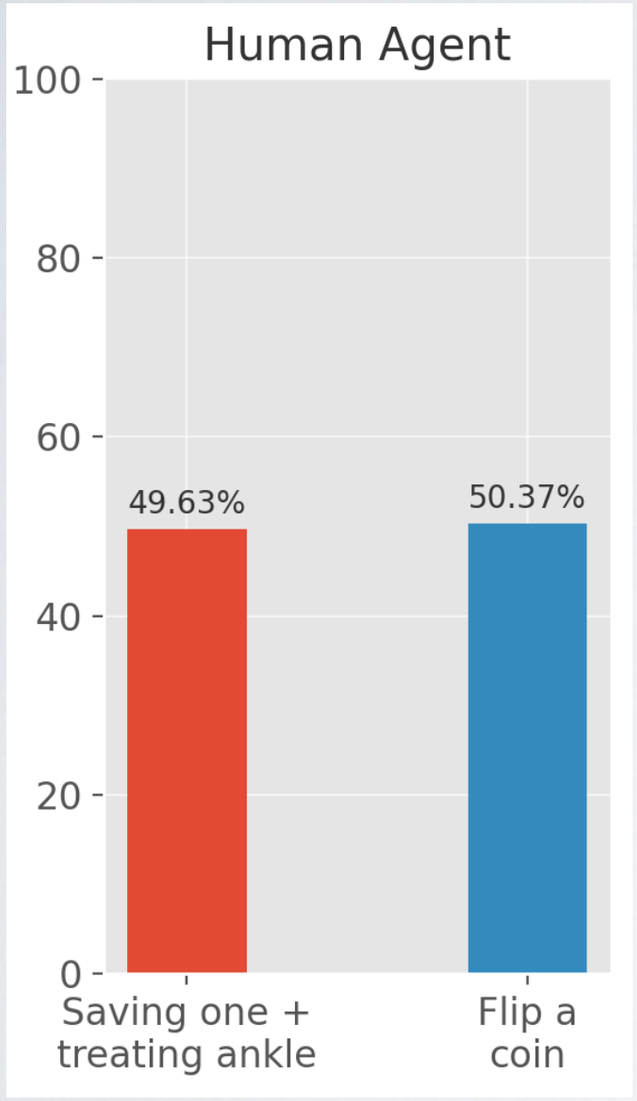


χ^2 (n=260)=13.75, p<.0001,
Cramer's V=.23 (medium-sized effect)



AI
Human OFF the loop

HUMAN/AI TEAMS



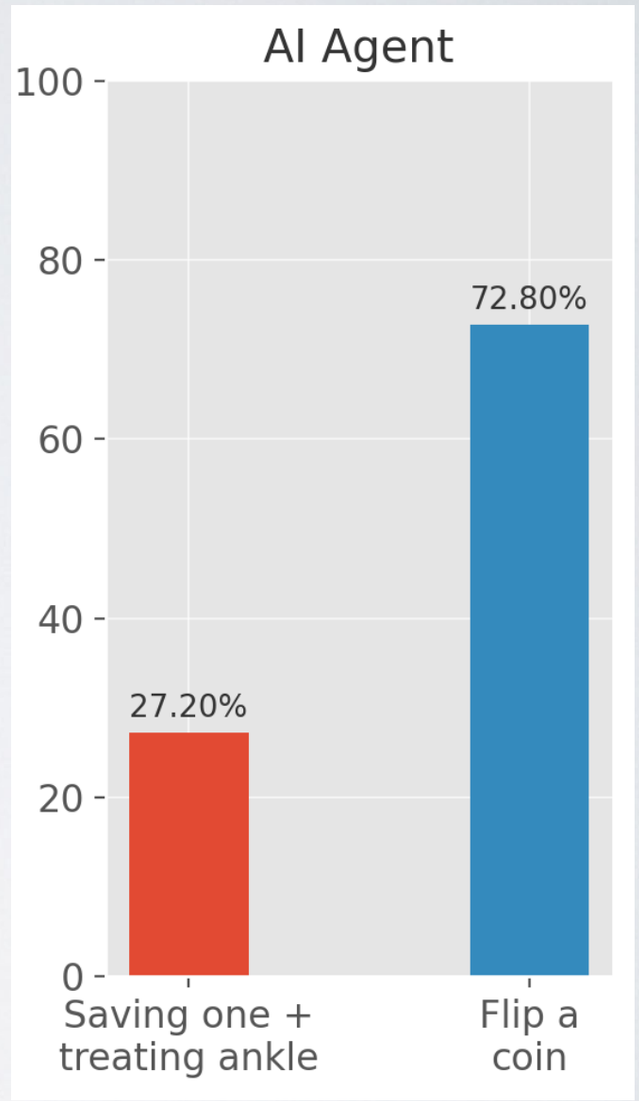
Human Agent

?

Team
Human IN the loop

?

Team
Human ON the loop



AI
Human OFF the loop

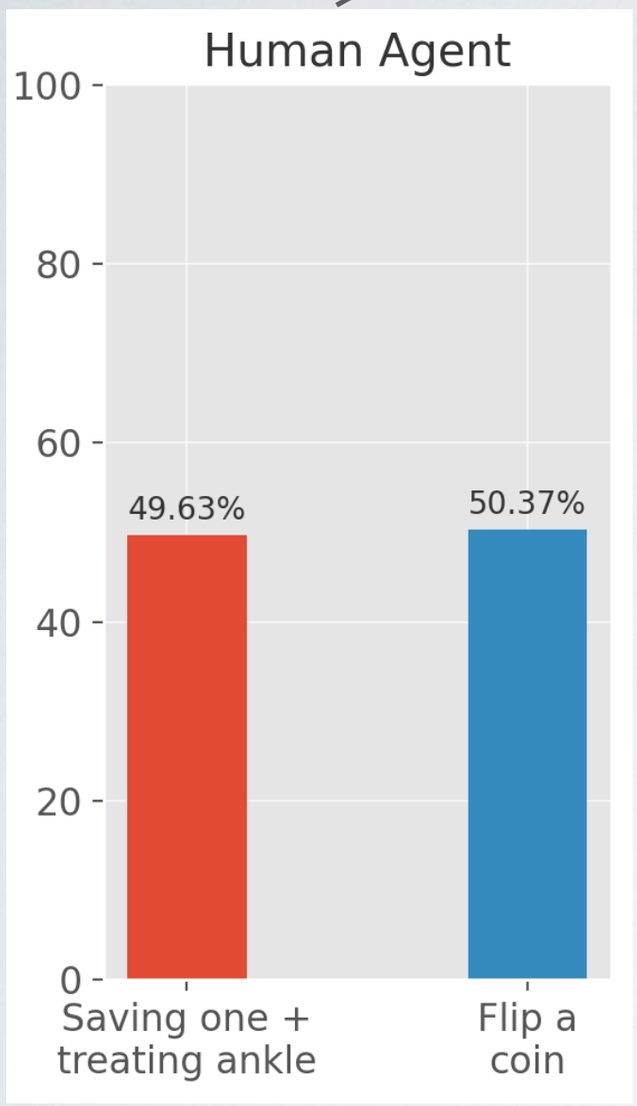
HUMAN/AI TEAMS

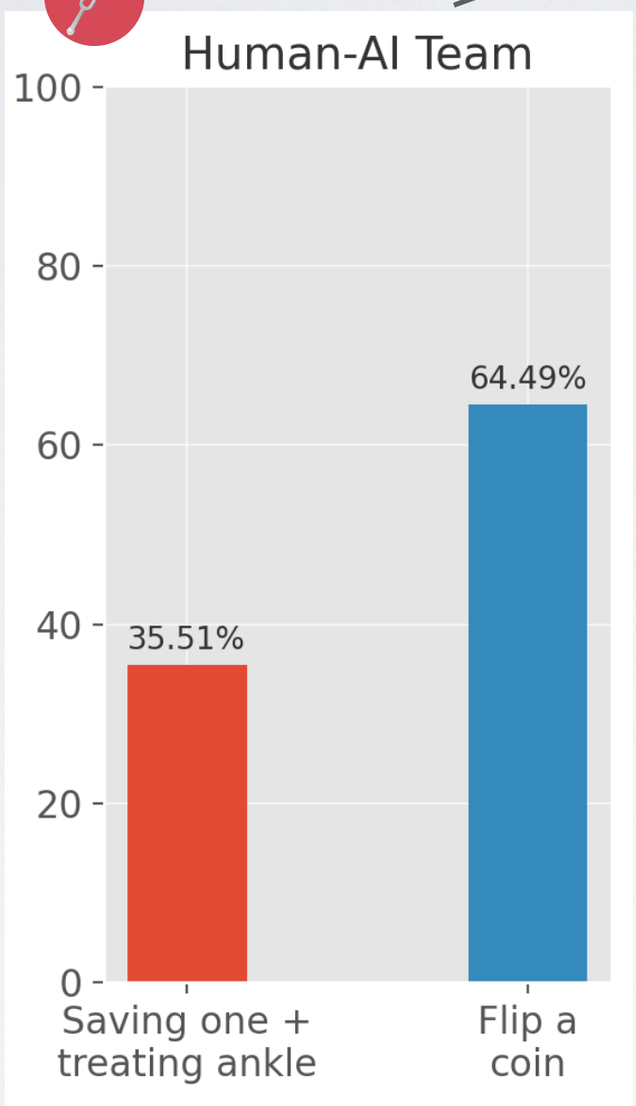
419 (459)

♀ 201 ∅ 42

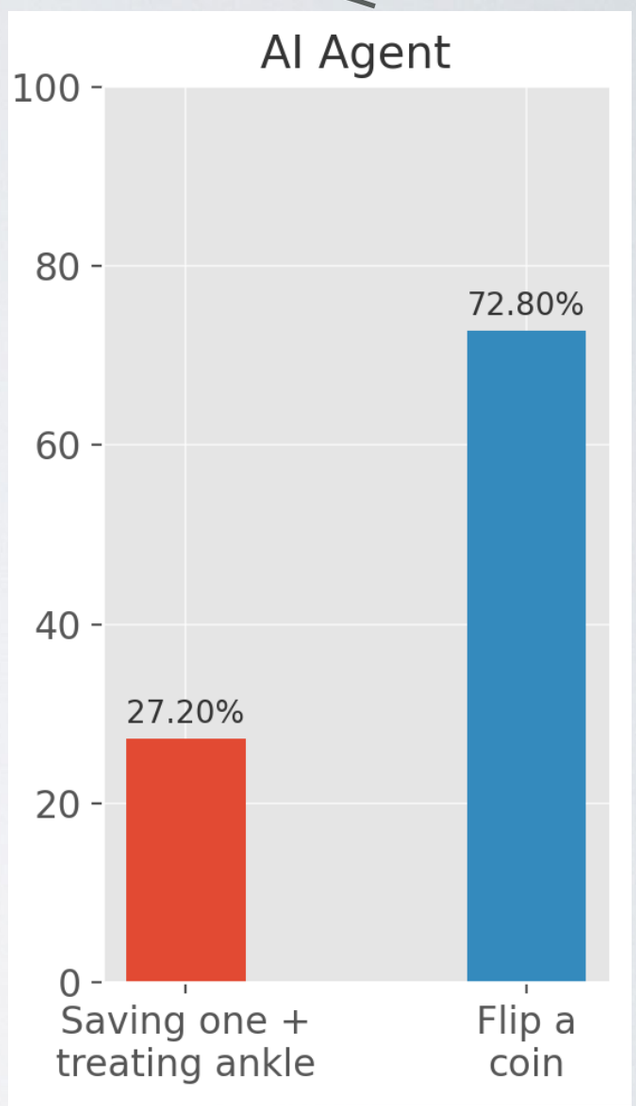
*

ns





?



Human Agent

Team
Human IN the loop

Team
Human ON the loop

AI
Human OFF the loop

Kneer (*in prep*): Value forks and human/AI decision-making

AI ALIGNMENT & VALUE FORKS

VALUE FORKS



- Value Forks arise if what is morally desirable/right/should be done depends on agent-type (human v. AI).

AI ALIGNMENT

- AI Alignment (Russell, 2019; Gabriel, 2021): Getting AI applications to produce outputs consistent with human values.
- For instance: Don't kill us.
- But also: Be a responsible conversational partner etc.

AI ALIGNMENT

- AI Alignment(Russell, 2019; Gabriel, 2021): Getting AI applications to produce outputs consistent with human values.
- Problem: value forks make alignment gets a lot more complex:
- Should we align AI to the values we expect a human to realize or to the values we expect AI to realize?
- And how will we judge the *human supervisor/alignment engineer* (who should do A) for the behavior of an AI agent (who should *not* do A)?

RECAP & TAKEAWAYS

KEY QUESTION

- Q: Are our traditional strategies for allocating responsibility fit for the world we live in?
- Philosophers: No. For instance: responsibility gaps.
- Psychologists: No: People blame machines and do other funny things.
- So we need *new* models: control architectures that allow the just allocation of responsibility.

NEW MODELS ?

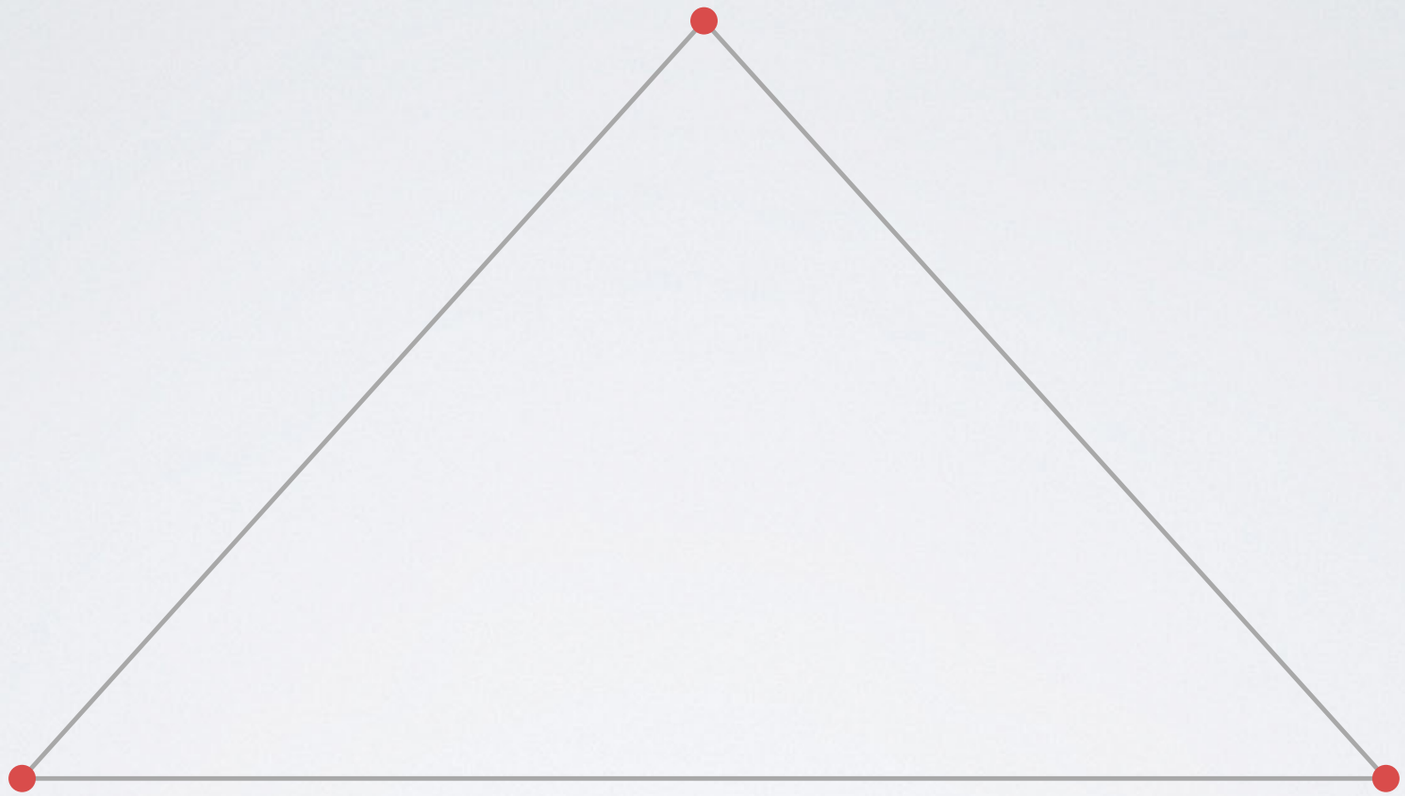
- HITL/HOTL (cf. e.g. EU AI act, Art. 14; GDPR, Art. 22)
  - People don't "feel it" (and neither will judges, juries, compliance depts.)
- Tracking & Tracing (De Sio & Van den Hoeven)
  - People don't "feel it" either
- Alignment
  - Complex (Value forks & other problems: What values? Who decides?)

NEW MODELS

- Still, we must urgently persist to work towards control architecturs and models of responsibility attribution in human-AI interaction which are:
  - Moral-philosophically adequate (and hence *just*)
  - Legally enforceable & practical
  - Psychologically intuitive

Ethics

Who is morally responsible?



Psychology

Whom do people consider responsible?

Law / Governance

Who is legally responsible?

If you are working on such models, let's collaborate.

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Slides & Refs.

THANK YOU.

Comments & critique welcome: hello@markuskneer.net

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